

- What kind of change takes place when dough is cooked into *chapati*?
 - Physical and reversible
 - Chemical and irreversible
 - Physical and irreversible
 - Reversible and temporary
- Which of the following best describes melting of ice?
 - Irreversible change
 - Chemical change
 - Reversible physical change
 - Natural chemical change
- Which one of the following changes results in a new substance being formed?
 - Freezing of water
 - Boiling the milk
 - Cutting a paper
 - Baking a cake
- Which process is involved in converting water ice cube into solid form?
 - Evaporation
 - Freezing
 - Sublimation
 - Infiltration

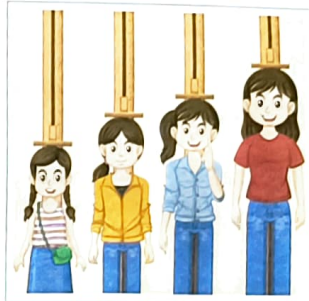


CHAPTER 02

Adolescence : A Stage of Growth and Change

Learning Outcomes...

- Growing with Age : The Teenage Years
 - Pubertal Changes
- Changes that Indicate Reproductive Capability
- Emotional and Behavioural Changes in Adolescents
- Making Adolescence a Joyful Experience
- Nutrition
- Personal Hygiene
- Physical Exercise
- Balanced Social Life
- Saying "No" to Harmful Substances
- The 'Why' Question for Adolescence



Introduction

The life of a human can be divided into different stages – infancy, childhood, adolescence, adulthood and old age. Adolescence is the transitional stage of human development, during which a juvenile matures into an adult. It is the phase in which physical and more subtle changes are noticed in humans, and it is a period of psychological and social transition from childhood to adulthood.

The World Health Organisation (WHO) defines adolescence as the period of life between 10 to 19 years of age. It is a period of transition from a childhood to an adulthood, i.e., the period of human development during teenage years. It is characterised by various physical, behavioral and emotional changes along with development of reproductive system. During adolescence, the body undergoes several changes. These changes mark the onset of puberty, i.e., the beginning of the body's journey to become an adult.

Growing with Age : The Teenage Years

Puberty is the stage of life during which an adolescent attains sexually maturity.

It occurs at the age of 9 to 12 years in girls and 10 to 14 years in boys. However, the duration may vary from person to person. Puberty ends when an adolescent reaches the reproductive maturity.

In males, the testes start producing sperms and secreting the male hormone testosterone. In females, the ovaries begin to release eggs and produce the female hormones estrogen and progesterone.

ACTIVITY CORNER

- To find out the common observable changes associated with adolescence
- **Procedure :** (i) Take a box and some paper slips.
(ii) Distribute the paper slips among the students of your class.
(iii) Ask students to write down the changes that they have observed in their body since last 3 years.
(iv) Now ask students to fold the slips and put them in the box.
(v) Based on the information of slips prepare a table enlisting the common changes observed in the body during that duration.

► Observation:

S.No.	Changes	Observations
1.	Height	Continuous increase in height.
2.	Weight and strength	Muscle growth is more prominent in boys than girls that lead to increase in weight as well as strength.
3.	Appearance	Some students have developed pimples on their skin. In some students, fine hair growth is observed on face and thick hair growth in armpits and pubic region.
4.	Changes in voice	In some boys the voice started to break and become hoarse and deep.

- **Conclusion :** There are many changes or natural signs that are associated with adolescence in both boys and girls. These changes are necessary for reproductive maturity.

Pubertal Changes

Certain changes are common in both boys and girls at puberty. The changes that takes place during puberty are known as **secondary sexual characteristics**. These traits helps distinguish males and females. These traits are discussed below:

1. Increase in height

During puberty, there is sudden increase in height due to the growth of bones of legs and arms. Initially, girls grow faster than boys, but by the age of 18 years, both boys and girls reach their maximum height. The rate of growth varies in different sexes (i.e., in boys and girls) and also in different individuals.

The height of an individual is a hereditary character. It depends on the genes inherited by an individual from parents. However, eating right type of food in right quantity during these growing years also influences the

height. Therefore, nutritional requirements of adolescent boys and girls are different from adults.

COMPETITION WINDOW

Changes in body shape

During puberty, boys and girls start looking different due to certain changes in their bodies.

In boys, the muscles of the body grow and become more prominent. Their shoulders become broad and chest becomes wide. The body becomes more muscular.

In girls, pelvic region widens, breasts develop and increases in size. Mammary glands, i.e., milk secreting glands develop inside the breasts.

Changes in voice box and voice

At puberty, the voice box or larynx begins to grow. In boys, it protrudes out in the throat region as Adam's apple. The larynx is smaller in girls than in boys and is hardly visible from outside. Boys develop deep voice (hoarse voice), while girls develop high pitched voice.

In some adolescent boys, the muscles of the growing voice box become out of control. This makes the voice hoarse and rough. Though, this is a temporary phase and voice returns to normal after sometime.

Adam's apple

Fig: The Adam's apple

Appearance of hair in different parts of body

Both boys and girls experience the growth of hair in different parts of their bodies, such as the armpits and pubic region. Boys often develop facial hair, which later grows into a moustache and a beard as they move towards adulthood. Some boys may also develop chest hair and back hair, though few boys may have no significant hair growth. Variations in timing of appearance of hair growth is completely normal.

Increased activity of sweat and sebaceous glands

At puberty, both sweat glands and sebaceous glands become more active especially on the face. This leads to appearance of acne and pimples on the face. This problem can be avoided by reducing intake of fat rich food and also keeping the face clean by washing it properly and repeatedly. Sweat glands are used to regulate temperature and remove waste by secreting water, sodium salts and nitrogenous waste such as urea onto the skin surface. The main electrolytes of sweat are sodium and chloride. Sweat is odourless but it smells due to bacteria present on our skin.

acne and pimples are made because of the oil from the sebaceous glands along the pores on the face. Don't forget!

DO YOU KNOW?

Sweat tastes salty due to the presence of sodium chloride in it.

Maturation of sex organs

At puberty, the male sex organs like testes and penis become fully developed in boys. In girls, the ovary enlargement, maturation and release of eggs begin which leads to the onset of menstrual cycle.

Changes that Indicate Reproductive Capability

Adolescence is marked not only by observable changes but also by internal changes that are not visible from the outside. One such change is the maturation of various parts involved in the process of reproduction. **In females, the reproductive age starts at puberty and continues till the age of 45 to 55 years.** The maturation and release of female gamete (the ovum) is cyclic, i.e., once in every 28 to 30 days. Many healthy girls may have longer or shorter menstrual cycles ranging from 21–35 days. The process of release of an ovum or egg by an ovary is called **ovulation**. During this period of 28–30 days, the wall of the uterus becomes thick so as to receive the fertilised egg. If the egg is fertilised by sperm, uterine wall develops and an embryo gets embedded in it which results in pregnancy.

In case fertilisation does not occur, the uterus lining and its blood vessels break down and pass out of the vagina in the form of blood and mucus along with the unfertilised egg. This bleeding in women is called **menstruation or period**. It lasts for 3 to 7 days.

These cyclic changes which take place in adolescent girls regularly every 28 days or so and is marked by menstrual flow is called a **menstrual cycle**. The menstrual cycle is an important natural process and is one of the signs of good reproductive health.

The first menstrual flow begins at puberty and is called **menarche**. At 45 to 55 years of age, the menstrual cycle stops. This stoppage of menstruation is termed as **menopause**.

Breaking myths about menstruation

There are many **wrong beliefs** about menstruation that often lead to unnecessary fear, shame, or even guilt. These beliefs have given rise to certain **myths and taboos**. Among such myths and taboos, physical isolation of menstruating girls is unfortunately still quite prevalent. Menstruation is a natural process, and such myths have no scientific basis. By **encouraging a scientific outlook** towards menstruation, we can help society appreciate the need to promote good reproductive health of women and contribute positively towards a **healthier lifestyle** for women.

Menstrual cycle consists of four phases:

- **Menstrual phase** (1st–5th day) : The wall of the uterus breaks down and menstruation begins.
- **Proliferative phase** (6th–13th day): Proliferation of the uterine wall occurs. The uterine wall becomes thicker by cell division.
- **Ovulatory phase** (14th day): Ovulation (release of ovum) occurs.

COMPETITION WINDOW

- **Secretory phase** (15th to 28th day) : Uterine glands produce increased amount of watery mucus. Such a uterine wall is necessary for the implantation of the fertilised ovum. In the absence of fertilisation, uterine wall breaks leading to menstruation marking a new cycle.

Illustrations

1 What is adolescence?

Ans.: The period in human development that marks the end of childhood and beginning of adulthood, when the body undergoes changes, leading to reproductive maturity is called adolescence.

2 What is ovulation?

Ans.: The process of release of an ovum or egg by an ovary during reproductive cycle of a female is known as ovulation. It usually occurs on the 14th day of menstrual cycle.

Emotional and Behavioural Changes in Adolescents

(When compared to childhood, adolescence is a phase that is marked by stronger emotions) These emotions have multiple effects on adolescent's behaviour. These changes may be exciting, confusing, or both. They may take up social initiatives to support the needy or might develop interest in a particular sector. Understanding how our emotions influence our behaviour and actions, can help us make better choices and respond to situations thoughtfully.

Mental, intellectual and emotional maturity

- Adolescence brings a change in a person's way of thinking.
- Adolescents want to be independent.
- Mental and physical changes taking place during adolescence cause mood swings.
- He/she finds a way to recognise and manage his/her emotions.
- Intellectual development takes place and they spend considerable time in thinking. The ability to see other's point of view, exploring ideas, developing concepts and memory skills improve.
- **Maturity in adolescents makes boys and girls interested in the opposite sex.**
- Sometimes, an adolescent may feel confused, self conscious or insecure while trying to adjust to the changes in the body and mind. This is a natural part of growing up and is experienced by everyone.

Making Adolescence a Joyful Experience

Health is a state of complete physical, mental and social well-being. A person is said to be enjoying good health when he is free from diseases, anxiety and tension. The journey of adolescence is a unique experience. Heightened curiosity and excitement during this stage of life give adolescents a new viewpoint towards almost everything around them. Good habits, thoughtful decisions, and small actions can have a powerful positive impact on the overall well-being of adolescents.

To keep the body healthy, an individual needs to have a balanced diet, personal hygiene and physical exercise. During adolescence, these needs become more essential as the body is growing. Any disease during adolescence period leads to retarded growth.

Nutrition

- Adolescence is a stage of rapid growth and development. Hence, during this stage balanced diet should be consumed. A balanced meal includes proteins, carbohydrates, fats and vitamins in requisite proportions. A diet consisting of a variety of healthy food, providing adequate amounts of nutrients necessary for good health is called **balanced diet**.

ACTIVITY CORNER



To study different healthy food sources and their role in growth and development

- **Procedure :** (i) Enlist the different sources of food that are good for your health.
(ii) Note down the nutrients present in them. You may take the help of books, internet and elders.

(iii) Write down all the information in an observation table.

► Observation :

Food Sources	Nutrients we get from them	Functions of these nutrients
Milk, millets, curd, cheese and paneer	Calcium, proteins, fats	Helps in optimal growth of bones and teeth
Fish, eggs, almonds and peas	Proteins	To help proper growth, gain strength, and improve levels of energy. Body needs proteins to make new tissues.
Spinach, kidney beans and dried fruits like raisins and figs	Iron	To help in the formation of blood.

- **Conclusion :** To support body growth, we need balanced diet so that we get all the required nutrients.

Vitamin B₁₂

Vitamin B₁₂ is an important nutrient needed for making red blood cells and keeping our nerves healthy. Since our body cannot make it, so it is obtained from the food we eat.

The main sources of Vitamin B₁₂ include:

- Animal-based foods:** milk, curd, cheese, eggs, fish, chicken, and meat.
- Fortified foods:** cereals, soy milk, nutritional yeast, etc.
- Supplements:** vitamin B₁₂ tablets or syrups (only if prescribed by a doctor).

Dorothy Hodgkin was a brilliant scientist who studied the structure of vitamin B₁₂. In 1964, she became the third woman to win the Nobel Prize in the field of chemistry.



Deficiency of one or more nutrients in our diet for a long time can lead to deficiency diseases and disorders.

Deficiency of Iron or Vitamin B₁₂ in Adolescents

Adolescents, especially girls, may sometimes suffer from blood-related health problems because of lack of iron or vitamin B₁₂ in the body. The most common problem is **anaemia**. Anaemia occurs when the blood does not have enough healthy red blood cells to carry oxygen to all parts of the body.

Symptoms:

- Anaemic person always feels tired or weak.
- They develop pale skin.
- Shortness of breath is frequent.
- Poor concentration or attention in class.

Managing Iron Deficiency:

We can manage iron deficiency through following ways:

- Eat iron-rich foods like green leafy vegetables, jaggery, dates, pulses, eggs, and iron-fortified cereals.
- Include Vitamin C-rich foods such as oranges, amla, and lemon to improve iron absorption.
- Avoid tea or coffee after meals.
- Take iron and folic acid tablets as advised by a doctor or school program.

Government Schemes to prevent Iron and Vitamin Deficiencies:

- Anaemia Mukht Bharat (Anaemia-Free India)** which aims to make people free from anaemia. They provide iron and folic acid tablets, health check-ups, and awareness programs in schools and communities.
- Weekly Iron and Folic Acid Supplementation (WIFS) Programme** which aims provide one iron-folic acid tablet every week to school students (boys and girls aged 10–19 years).
- Mid-Day Meal Scheme and Poshan Abhiyaan** provide nutritious and balanced meals to school children. It focus on including foods rich in iron and vitamins to prevent malnutrition and anaemia.

Personal Hygiene

It is very important for the teenagers to take care of their cleanliness. Maintaining body hygiene, particularly in the armpits and the pubic region, may protect us from potential infections. There is an increased activity of sweat glands which makes the body smelly. Hence, bathing everyday is very important. If cleanliness is not maintained, then there are chances of catching infections. For girls, maintaining proper hygiene during menstruation is essential for both comfort and health. During menstrual flow, girls should take special care of cleanliness, otherwise they can get affected by various RTIs (Reproductive Tract Infections). Menstrual hygiene can be maintained by using items such as sanitary pads and reusable cloth pads designed for this purpose. Used sanitary pads should be wrapped in paper and disposed of in a dustbin to keep the surroundings clean and prevent the spread of infections. Nowadays biodegradable sanitary pads are available, which are environment-friendly and help in maintaining cleanliness and hygiene.

To support menstrual hygiene, the government has introduced several initiatives:

- Menstrual Hygiene Scheme (MHS):** Under this scheme by the Government of India, sanitary pads are provided free of charge or at a reduced price to adolescent girls in rural areas. Awareness programmes are also conducted to educate girls about menstrual hygiene and health.

- **Rashtriya Kishor Swasthya Karyakram (RKSK):** It aims to improve the overall health of adolescents, including menstrual health and hygiene. It encourages peer education, where older students help younger ones learn about these topics.
- **Suvidha Sanitary Napkin Initiative:** Under this initiative, biodegradable sanitary pads are provided at affordable prices through Jan Aushadhi Kendras. The initiative aims to improve access to menstrual hygiene products for women and adolescent girls.
- **State-Level Initiatives:** Various state governments have their own programmes, such as the 'Shuchi Scheme' in Karnataka and free sanitary napkin schemes in states like Tamil Nadu and Odisha. These programmes aim to distribute free sanitary pads in government schools.

Physical Exercise

- It is very important for teenagers to walk and exercise as it keeps the body fit and healthy. It also builds up stamina and boosts mood.



Fig.: Child jumping rope



Fig.: Children doing yoga

- Physical activities keep our body healthy and strong. They help in proper growth, improve blood circulation, and make our muscles and bones stronger. Regular exercise or outdoor games also refresh the mind, reduce stress, and keep us active throughout the day.

Balanced Social Life

- Since adolescence is a stage of life marked by new experiences and emotions, we must be thoughtful and responsible while interacting with others – whether in person or through social media. It is the time when adolescents may feel attracted to their peers, and mimic their behaviour. Modern technology has made online platforms easily accessible for everyone to gather information, connect, interact, and share ideas. We must use these platforms wisely and responsibly for the well-being of all. At times, people may use social media carelessly or without awareness, which can lead to problems. Taking guidance from elders and teachers helps us use these platforms safely and positively.

Cyberbullying refers to the use of digital devices such as phones, computers, or online platforms to harass others by sending misleading messages, spreading false rumours, or sharing personal information without consent.

Saying "No" to Harmful Substances

- Since adolescence is also a stage of insecurity and confusion, it is easy for the adolescent mind to get influenced by wrong company or advice and fall prey to drug and alcohol abuse. Some people start consuming harmful substances such as tobacco, *gutka*, cigarettes, *beedi*, alcohol, etc.
- Drugs are very addictive and if once taken, then there is a tendency to take them again and again. This is called as substance abuse. These drugs harm the body in the long run, ruining health and happiness. These harmful substances can cause serious health problems, such as breathing difficulties, memory loss and damage to the lungs, depending on the substance.
- To stay healthy, it is important to completely avoid these substances and make healthy choices instead. It is important to remain firm and confident in your decision to say 'NO'. The important steps to overcome addiction are as follows:
 - Talk to parents, teachers, or a counsellor for support and guidance.
 - Involve yourself in studies, hobbies, sports, or creative activities to divert your mind.
 - Stay away from people, places, or situations that encourage the addiction.
 - Take one step at a time toward giving up the habit.
 - Believe in yourself and celebrate small improvements.
 - Doctors or counsellors can provide proper advice and treatment if required.

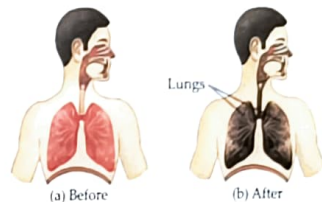


Fig.: Condition of the lungs before and after prolonged exposure to bidi/cigarette smoke

Nasha Mukta Bharat Abhiyaan

The Nasha Mukta Bharat Abhiyaan was launched by the Ministry of Social Justice and Empowerment, Government of India. It intends to reach out to the masses and spread awareness on the issue of avoiding substance abuse through active participation of the youth, women, and the community. The special focus is on the prevention of substance abuse among early-age children.

The Government has set up a National De-addiction Helpline (14446) to provide support and guidance to individuals struggling with drug addiction and to help them recover.

The 'Why' Question for Adolescence

Many changes in adolescence, including menstruation and other signs of puberty, are mainly due to **hormones**—certain chemicals produced in our bodies. Hormones play a crucial role in regulating various aspects of growth and development, contributing to the proper functioning of the body. These are produced in different parts of the body and are released at an appropriate time in response to signals from the brain. Some hormones influence mood and behaviour.

The hormonal control in the development of secondary sexual characters in boys and girls at puberty is by testes and ovaries that release testosterone in males and estrogen in females respectively and they are called as sex hormones.

Illustrations

- 3 Name the hormones responsible for changes during adolescence.

Ans.: Testosterone in boys and estrogen in girls are responsible for bringing changes during adolescence.

- 4 Why adolescents should take balanced diet?

Ans.: Adolescence is a stage of rapid growth and development. Hence, during this stage balanced diet should be consumed. A diet consisting of different types of food, providing adequate amounts of the nutrients necessary for good health is called balanced diet. Balanced diet includes proteins, carbohydrates, fats and vitamins in requisite proportions.

- 5 Name and explain the term associated with the misuse of social media platform.

Ans.: Cyberbullying involves the use of digital devices such as phones, computers on online platforms to harass others by sending misleading messages, spreading false rumours, or sharing personal information without consent.

- 6 What is the significance of modern technologies in the social life of an adolescent?

Ans.: Modern technologies have made online platforms available for all to obtain information, make connections, interact and share information with each other. These platforms should be used responsibly. Elders and teachers can play a crucial role in guiding adolescents to use these platforms positively.



EVERYDAY LIFE EXAMPLES

Adolescence is a fascinating and dynamic period of life, serving as the crucial bridge between childhood and adulthood. It is defined by rapid and profound changes that affect nearly every aspect of a person's being—physical, emotional, social, and behavioral.

Some real-life examples of key concepts related to adolescence are:

- ◆ **Mood swings:** One day, a 12-year-old girl might be extremely excited about a school project, but the next morning, she might feel upset and irritable about something minor, like her favourite shirt being dirty. Hormones can cause these strong and rapid changing feelings.
- ◆ **Search for identity:** A teenager might suddenly develop a new interest in a specific sport, music genre, or social cause (like an environmental group), trying out different interests to see what fits their personality and who they are becoming.
- ◆ **Developing social responsibility:** A group of 15-year-olds might decide to organise a cleanliness drive in their local park, showing a newfound interest in their community and social well-being beyond their immediate needs.
- ◆ **Decision making:** An adolescent might start weighing the pros and cons of joining an extra-curricular activity versus focusing solely on studies, using more complex problem-solving skills than before.

CONCEPT MAP

Adolescence : A Stage of Growth and Change

Adolescence

- A period in human development that occurs between childhood and adulthood (10-19 years).
- During this period, body undergoes changes leading to reproductive maturity.

Puberty

Period of life during which an individual becomes sexually mature.

- Changes at puberty
- Increase in height
 - Growth of muscles (more in boys than in girls)
 - Change in body shape
 - Change in voice
 - Increased activity of sweat and sebaceous glands
 - Maturation of sex organs
 - Mental, social and emotional maturity
 - Appearance of secondary sexual characters

In boys

- Beard, moustaches
- Pubic hair
- Hair under armpits and on chest
- Adam's apple
- Deep voice

Menstrual cycle

- These are cyclic changes which take place in females (reproductive system) regularly every 28 days. It is marked by menstrual flow.
- The first menstrual flow begins at puberty and is called menarche.
- The stoppage of menstruation at 45 to 50 years of age is termed as menopause.

In girls

- High pitched voice
- Pubic hair
- Hair under armpits
- Development of breasts
- Broadening of pelvis

Making adolescence a joyful experience

- Heightened curiosity and excitement during this stage of life give adolescents a new viewpoint.
- Good habits, thoughtful decisions, and small actions have positive impact.
- During adolescence, there are five important necessities:
 - Nutrition
 - Personal hygiene
 - Physical exercise
 - Balanced social life
 - Saying "no" to harmful substances

Why adolescence?

- Many changes in adolescence occur due to hormones (chemicals produced in our bodies).
- Testosterone in males and estrogen in females control adolescent characters.
- Hormones regulate various aspects of growth and development, contributing to the proper functioning of the body.

Emotional and behavioural changes in adolescents

- Mood swings
- Strong emotions
- Compassion
- Increased sensitivity
- Engage in music, dance or sports
- Self-exploration and involvement in creation and innovation, etc.

Solved Examples

1. Why do acne and pimples appear during adolescent period?

Ans.: During adolescent period, both sweat glands and sebaceous glands become more active especially on the face. This leads to appearance of acne and pimples on the face.

2. Define cyberbullying.

Ans.: Cyberbullying is the use of digital devices such as phones, computers on online platforms to harass others by sending misleading messages, spreading false rumours, or sharing personal information without consent.

3. What marks the end of adolescence?

Ans.: Reproductive maturity marks the end of adolescence.

4. What is Adam's apple?

Ans.: The voice box or larynx which protrudes out in the throat region during puberty in boys is known as Adam's apple.

5. Name the chemicals responsible for changes taking place in the body.

Ans.: Hormones

6. Mention any two distinguishing features that are observed in boys and girls at puberty.

Ans.: Two features seen in boys at puberty are:

- Growth of facial hair.
- Deepening of voice due to enlargement of voice box, seen as a protruding part of the throat called Adam's apple.

Two features seen in girls at puberty are:

- Development of breast.
- Widening of the region below the waist.

7. Define adolescence.

Ans.: The period of life, when the body undergoes changes, leading to reproductive maturity is called adolescence.

8. What personal hygiene should be maintained by the teenagers?

Ans.: It is very important for the teenagers to take care of their cleanliness as there is an increased activity of sweat glands which makes the body more prone to bacterial infection and smelly. Hence, bathing everyday is very important. Cleanliness is not maintained, there are chances of catching infections. During menstrual flow, girls should take special care of cleanliness, otherwise they can get affected by various RTIs (Reproductive Tract Infections). It is advised to use good sanitary pads or reusable cloth pads at that time.

9. What is menstruation? Explain.

Ans.: When egg released by ovary does not get fertilised, it along with the thickened lining of the uterus and its blood vessels shed out of the body as bleeding in woman. This bleeding is called menstruation.

10. John and Radha were classmates since childhood. When Radha became eleven years old, she developed a little swelling on her neck. She visited the doctor who started medication for her. After a few years, John also developed a slight protrusion on his throat. He got worried and went to the doctor. But, the doctor assured him that it was a normal feature in boys while they are growing up. Can you think of any reasons for the difference in diagnosis?

Ans.: The swelling on Radha's neck could be due to goitre, a disease of the thyroid gland during which the gland produces insufficient quantity of thyroxine hormone. However, the protrusion in John's throat is due to the growth of voice box called Adam's apple which is formed during puberty in adolescent boys.

NCERT Section

1. Ramesh, an 11-year-old boy, developed a few pimples on his face. His mother told him that this is because of ongoing biological changes in his body.

- What could be the possible reasons for the development of these pimples on his face?
- What can he do to get some relief from these pimples?

2. Which of the following food groups would be a better option for adolescents and why?



(i)



(ii)

3. Unscramble the underlined word in the following sentences:

- The discharge of blood in adolescent girls which generally occurs every 28–30 days is ntsmnoiaertu.
- The hoarseness in the voice of adolescent boys is due to enlarged iccoov xob.
- Secondary sexual characteristics are natural signs that the body is preparing for adulthood and mark the onset of urtypeb.
- We should say NO to lahoclo and srugd as they are addictive.

4. Shalu told her friend, "Adolescence brings only physical changes, like growing taller or developing body hair." Is she correct? What would you change in this description of adolescence?

5. During a discussion in the class, some of the students raised the following points. What questions would you ask them to check the correctness of these points?

- Adolescents do not need to worry about behavioural changes.
- If someone tries a harmful substance once, they can stop anytime they want.

6. Adolescents sometimes experience mood swings. On some days, they feel very energetic and happy, while on other days, they may feel low. What other behavioural changes are associated with this age?

7. While using a toilet, Mohini noticed that used sanitary pads were scattered near the bin. She got upset and shared her feelings with her friends. They discussed the importance of menstrual hygiene and healthy sanitary habits. What menstrual hygiene and sanitary habits would you suggest to your friends?

8. Mary and Manoj were classmates and good friends. On turning 11, Mary developed a little bulge on the front of her neck. She visited the doctor who gave her medication and asked to take iodine-rich diet. Similarly, a bump was developed on the front of Manoj's neck when he turned 12.

However, the doctor told him that it was a part of growing up. According to you, what could be the possible reason for advising Mary and Manoj differently?

9. During adolescence, the boys and girls undergo certain physical changes, a few of which are given below.

- Change in voice
 - Development of breasts
 - Growth of moustache
 - Growth of facial hair
 - Pimples on the face
 - Growth of hair in the pubic region
 - Growth of hair in armpits
- Categorise these changes in the table given below:

Physical changes during adolescence		
Observed only in boys	Observed only in girls	Common in boys and girls

Exercise

Multiple Choice Questions

LEVEL - 1

Growing with Age : The Teenage Years

- Adolescents are also known as
 - adults
 - Teenagers
 - infants
 - fetus.

(NCERT Page 74)
- Adam's apple is due to
 - inflammation of mammary glands
 - enlarged voice box
 - swelling in voice box
 - atrophied thyroid gland.

(Page 76)
- Why do girls grow faster than boys at the age of 12 years?
 - Girls reach puberty earlier than boys.
 - Boys reach puberty earlier than girls.
 - Girls normally take in more proteins at the age of 12 years.
 - Boys normally consume less food.

(U)
- Protruding voice box found in boys is known as _____.
 - pharynx
 - larynx
 - adam's apple
 - both (b) or (c).

(Page 76)
- Which of the following is/are secondary sexual characteristic(s) of a boy?
 - Growth of facial hair
 - Bones and muscles development
 - Growth of pubic hair
 - All of these

(Page 77)
- Secondary sexual characteristics in girls usually develop between the age of
 - 6 to 8 years
 - 9 to 12 years
 - 15 to 18 years
 - 19 to 21 years.

(Page 77)

- The changes taking place during adolescence are the signs of
 - puberty
 - adulthood
 - childhood
 - abnormal growth.

(Page 77)
- Which of the following is not a secondary sexual characteristic of a girl?
 - Enlargement of the breasts
 - Deepening of voice
 - Onset of menstruation
 - Growth of hair in armpit and pubic region

(Page 77)
- Adolescence is a phase marked by
 - physical changes
 - hormonal changes
 - emotional changes
 - all of these.
- Adolescence marks the onset of which phase?
 - Puberty
 - Menstrual
 - Adult
 - Childhood

(Page 77)

Changes that Indicate Reproductive Capability and Emotional and Behavioural Changes in Adolescents

- Each menstrual cycle usually lasts for about _____ days.
 - 7
 - 14
 - 21
 - 28

(Page 77)
- The monthly discharge of blood from the uterus via vagina in females is called
 - ovulation
 - menstruation
 - fertilisation
 - implantation.

(R) (Page 77)

- Reproductive age in woman starts when their
 - menstruation starts
 - breasts start developing
 - body weight increases
 - height increases.

(R) (Page 77)
- Menstruation will not take place if
 - the ovum is released
 - fertilisation does not occur
 - the hormones are produced
 - fertilisation occurs.

(U)
- Which of the following is the common symptom of menstruation?
 - Heart pain
 - Abdominal pain
 - Nosebleed
 - Broken bones

(Page 77)
- Menstrual cycle is a sign of
 - bad reproductive health
 - good reproductive health
 - abnormal activity of gonads
 - none of these

(U) (Page 77)
- Which of the following changes causes mood swings during adolescence?
 - Mental changes
 - Physical changes
 - Changes in voice box
 - Both (a) and (b).

(R) (Page 78)

Making Adolescence a Joyful Experience

- Adolescents should be careful about what they eat, because
 - proper diet develops their voice box
 - proper diet is needed for the rapid growth taking place in their body
 - adolescents feel hungry all the time
 - taste buds are well developed in teenagers.

(Ap) (Page 79)
- The right meal for adolescents consists of
 - chips, noodles and coke
 - chapati, dal and vegetables
 - rice, noodles and burger
 - vegetable cutlets, chips and lemon drink.
- Which condition will result from iron in the bodies of adolescents?
 - Rickets
 - Asthma
 - Anaemia
 - High blood sugar
- At puberty, a diet rich in protein is for
 - supplying sufficient energy
 - the growth and repair of body
 - the formation of strong bones and teeth
 - protection from diseases.
- Rishi often drinks milk and eats leafy vegetables daily. Which nutrient is he mainly getting that helps build bones and teeth?
 - Iron
 - Calcium
 - Vitamin C
 - Iodine
- What is the function of balanced diet on body?
 - Rapid growth and development
 - Provide adequate amount of nutrients
 - Both (a) and (b)
 - Decreases appetite
- A Government programme that provides iron and folic acid tablets, health checkups aiming to make people anaemia free is known as
 - Weekly Iron and Folic Acid Supplementation (WIFS) Programme
 - Anaemia Mukh Bharat (Anaemia-Free India)
 - Mid-Day Meal Scheme and Poshan Abhiyaan
 - Rashtriya Kishor Swasthya Karyakram (RKSK)

25. Which among the following is the scientific contribution of Dorothy Hodgkin?

- (a) Invented the microscope
- (b) Studied the structure of vitamin B₁₂
- (c) Discovered carbohydrates
- (d) Discovered DNA

(Page 80)

26. What does RKSK stands for?

- (a) Rashtriya Krishi Swasthya Karyakram
- (b) Rashtriya Kishor Swasthya Karyakram
- (c) Rashtriya Kishor Swasthya Karyalaya
- (d) Rashtriya Krishi Samasth Karyalaya

(Page 81)

The 'Why' Question for Adolescence

27. Testes secrete

- (a) thyroxine
- (b) estrogen
- (c) testosterone
- (d) iodine.

28. Changes during puberty in females are caused by

- (a) sex cells
- (b) enzymes
- (c) nutrients
- (d) sex hormones.

29. Which of the following refers to female hormone?

- (a) Insulin
- (b) Estrogen
- (c) Testosterone
- (d) Both (a) and (c).

30. Which hormone is responsible for pubertal changes in males?

- (a) Testosterone
- (b) Estrogen
- (c) Insulin
- (d) None of these

31. Which of the following is responsible for reproductive growth of the body?

- (a) Hormones
- (b) Minerals
- (c) Ions
- (d) Plasma

(Page 84)

32. Hormones influence

- (a) mood
- (b) growth
- (c) behaviour
- (d) all of these.

(Page 84)

33. Which of the following is the main objective of the Nasha Mukta Bharat Abhiyaan?

- (a) Increasing tobacco sales and reducing alcohol sales.

- (b) Eliminating substance abuse in India
- (c) Promoting gaming culture.
- (d) Maintaining proper hygiene and sanitation.

(Page 84)

LEVEL - 2

34. At about 14th day of the menstrual cycle, a mature egg is released from an ovary. This is known as

- (a) attraction
- (b) fertilisation
- (c) menstruation
- (d) ovulation.

35. Which of the following statements are correct regarding the menstrual cycle?

- (i) Unfertilised ovum is released during menstrual cycle.
- (ii) The uterine lining becomes thin immediately after menstruation.
- (iii) Physical isolation of menstruating girls is very essential for maintaining the health of others.

- (a) (i) and (ii)
- (b) (i) and (iii)
- (c) (ii) and (iii)
- (d) (i), (ii) and (iii)

36. Which of the following statements is/are true?

- (a) Boys reach puberty at an earlier age than girls.
- (b) At about the age of 12 years, girls grow more rapidly than boys.
- (c) Between the age of 4 to 10 years, girls grow faster than boys.
- (d) All of these

37. The correct age at which the rapid growth occurs in male and female is

- | Male | Female |
|-----------------|-------------------|
| (a) 10-14 years | 9-12 years |
| (b) 0-3 years | 18 years onwards |
| (c) 12-14 years | 14 years onwards |
| (d) 4-12 years | 14 years onwards. |

38. Which of the following will occur in females at puberty?

- (i) Menstruation
- (ii) Enlargement of breasts
- (iii) Broadening of hips
- (a) (i) and (ii) only
- (b) (i) and (iii) only
- (c) (ii) and (iii) only
- (d) (i), (ii) and (iii)

39. What will happen if personal hygiene is not considered during menstruation?

- (a) Reproductive tract infections
- (b) Cramps
- (c) Infertility
- (d) Pregnancy

40. A girl has menstruation on the 6th day of the month. When is ovulation most likely to occur?

- (a) 15th - 17th day
- (b) 18th - 20th day
- (c) 23rd - 25th day
- (d) 27th - 28th day

41. Which of the following changes are common in both boys and girls?

- (a) Increase in height
- (b) Both sweat and sebaceous glands become more active
- (c) Growth of moustache and beard
- (d) Both (a) and (b)

42. Choose the incorrect statement from the following.

- (a) The height of an individual is a hereditary character.
- (b) Eating right type of food in right quantity during growing years also influences the height.
- (c) Nutritional requirements of adolescent boys and girls are same as that of adults.
- (d) None of these

43. Which of the following is not a characteristic change associated with adolescent boys?

- (a) Growth of muscles in the body

- (b) Shoulders become broad and chest become wide
- (c) Region below the waist becomes wider
- (d) Voice box or larynx begins to grow

44. Incomplete development of male secondary sexual characteristics is due to under secretion of the hormone

- (a) estrogen
- (b) progesterone
- (c) adrenaline
- (d) testosterone.

45. Problem of acne and pimples can be avoided by

- (a) reducing intake of junk foods
- (b) keeping the face clean by washing it properly and repeatedly
- (c) increasing intake of junk foods.
- (d) both (a) and (b)

46. A person goes through the following stages of growth. Which is the correct sequence?

- | | |
|-------------------------|-------------------------|
| 1. Fetus | 2. Zygote |
| 3. Embryo | 4. Baby |
| 5. Adult | 6. Adolescent |
| 7. Child | |
| (a) 2, 3, 1, 4, 7, 6, 5 | (b) 2, 1, 3, 4, 7, 5, 6 |
| (c) 3, 2, 1, 4, 6, 7, 5 | (d) 1, 3, 2, 4, 7, 6, 5 |

47. The 'Shuchi scheme' is implemented in which Indian state?

- (a) Odisha
- (b) Uttar Pradesh
- (c) Tamil Nadu
- (d) Gujarat

48. Following are the events that occur during menstrual cycle. Which is the correct sequence?

- (i) The uterus wall thickened with blood vessels.
- (ii) Discharge of blood from the body.
- (iii) The uterus wall breaks down.
- (iv) The ovary discharges an ovum.
- (a) (i), (iii), (ii), (iv)
- (b) (ii), (iv), (i), (iii)
- (c) (iii), (i), (iv), (ii)
- (d) (iv), (ii), (iii), (i)

49. What are the government efforts in menstrual hygiene?

- Educating girls about menstruation.
 - Distributing free/reduced cost pads
 - Selling pads at low cost
- (a) I and II (b) II and III
(c) I and III (d) All of these

50. What is drug abuse?

- Taking harmful substances or using medicine without a doctor's advice.
- Taking medicine after a doctor's advice.
- Using drugs only for medical treatment.
- Avoiding all kinds of drugs.

LEVEL - 3 (HOTS)

Competency Focused Questions (CFQs)

51. Read the given paragraph where few words have been italicised.

Secondary sexual characteristics are the traits that distinguish the two sexes of an individual. In human beings, secondary sexual characteristics in males include gain in height, a *reduced* Adam's apple, deep voice, growth of beard, moustaches and *pubic hair*, hair on the chest, thighs and armpits. In females, the secondary sexual characteristics are seen as height gain, but with shorter stature than boys, *more* facial hair, growth of pubic hair, hair in armpits, *high-pitched* voice and development of breasts. Other features include *shortening* of pelvis and deposition of fat around hips and *smoother* skin texture.

Select the correct option regarding these.

- Reduced* should be replaced by *prominent*.
- High-pitched* should be replaced with *low-pitched*.
- More* and *shortening* should not be replaced as they are correctly mentioned.
- Smoother* must be replaced by *rougher*.

52. Read the following statements and select the option which correctly fills the blank in any two of these statements.

- _____ is the period of life at which sexual immature boy or girl becomes sexually mature and capable of reproduction.
 - Sweat glands are responsible for the regulation of _____.
 - Menstruation lasts for _____ to _____ days.
 - Sex hormones are secreted by _____ and _____.
- (a) (i) Puberty, (iv) testes, ovary
(b) (ii) height, (iii) 10, 15
(c) (iii) 3, 7 (iv) pituitary, hypothalamus
(d) (ii) temperature, (iv) kidney, small intestine

53. A boy named Vansh who is studying in class 7, is facing the problem of pimple and acne on his face. He is quite worried about the changes that are taking place in his body. Given below are some of the preventive measures. Select the correct preventive measures which help Vansh to overcome his problem.

- Reduce intake of fatty foods.
 - Isolate himself from others.
 - Keep the face clean by washing it properly.
 - Drink minimal amount of water.
- (a) Only I (b) Only I and II
(c) Only I and III (d) Only IV

54. Abhinav is a 15 years old boy. He was quite well in his studies during middle school, but he is facing a sharp academic decline in high school exams. Due to this, his peers pressured him to try smoking and substance abuse. Which of the following should be followed by Abhinav in the current situation?

- He should follow the advice given by his peers.

- He should refuse firmly and walk away.
- He should discuss the problem with parents.

- He should try smoking at once to feel free.

- (a) Only I and III (b) Only II and III
(c) Only I and IV (d) Only II and IV

55. Read the given statements and select the correct option.

Statement I : Menstruation is a sign of weakness.

Statement II : Societal stigma around menstruation have negative impacts on mental and physical health.

- Both statements I and II are correct.
- Statement I is correct but statement II is incorrect.
- Statement I is incorrect but statement II is correct.
- Both statements I and II are incorrect.

Fill in the Blanks

- Changes that are seen during adolescence are primarily controlled by _____.
- Sweat is odourless but it gains odour when it mixes with the _____ present on our skin.
- The process of release of an ovum or egg by an ovary is known as _____.
- Menstruation or period lasts for _____ days.
- _____ scheme in Karnataka aims to provide free sanitary pads in government schools.
- A diet consisting of a variety of different types of food and providing adequate amounts of the nutrients necessary for good health is called _____.
- A balanced meal includes _____, _____, _____ and _____ in requisite proportions.
- Drugs are very _____ and if once taken, then there is a tendency to take them again and again.

- Adolescence is a period of psychological and social transition from childhood to _____.
- At puberty _____ and _____ glands become more active specially on the face.

True or False

- Testes and ovaries release estrogen in males and testosterone in females.
- In females, the reproductive age starts at puberty and continues to the age of 45 to 55 years.
- The first menstrual flow begins at puberty and is called menopause.
- Adolescents always want to be dependent on their parents.
- To ensure community health, used sanitary pads should be disposed inside a dustbin.
- Adolescence is a stage of life marked by new experiences and emotions.
- Cyberbullying involves the use of digital platforms to harass a person.
- The rate of growth is similar in different sexes and also in different individuals.
- Nutritional requirements of adolescent boys and girls are same as of adults.
- Spinach and dried raisins are a good source of iron.

Assertion & Reason Type

Directions : In the following questions, a statement of assertion is followed by a statement of reason. Mark the correct choice as :

- If both assertion and reason are true and reason is the correct explanation of assertion.

(b) If both assertion and reason are true but reason is not the correct explanation of assertion.

(c) If assertion is true but reason is false.

(d) If assertion is false but reason is true.

1. **Assertion** : Adolescence is a critical phase of human development, during which a juvenile matures into an adult.
Reason : Adolescence is the phase of physical, psychological and social changes.

2. **Assertion** : Puberty is the age at which an individual becomes sexually mature.

Reason : In girls, puberty occurs at younger age than boys.

3. **Assertion** : The stoppage of menstruation is called menarche.

Reason : At 45 to 50 years, the menstrual cycle stops.

4. **Assertion** : In females, ovaries produce female hormone or estrogen.

Reason : Estrogen induces puberty changes in females.

5. **Assertion** : Calcium helps in the formation and strengthening of bones.

Reason : Milk, cheese and curd are a good source of calcium.

6. **Assertion** : To keep the body healthy, an individual need to have a balanced diet, personal hygiene and physical exercise.

Reason : During adolescence, these needs are not necessary.

7. **Assertion** : Teenagers need to take care of their cleanliness.

Reason : During adolescence, there is an increased activity of sweat glands which makes the body smelly.

8. **Assertion** : Adolescence is a stage of mature and stable mind.

Reason : Adolescent minds are easily influenced by wrong company or advice.

9. **Assertion** : During puberty there is a sudden increase in height of boys and girls.
Reason : The sudden growth occurs due to the growth in long bones of arms and legs.

10. **Assertion** : The larynx is smaller in boys than in girls and is hardly visible from outside.

Reason : Boys develop deep voice (hoarse voice) while girls develop high pitched sweet voice.

Subjective Questions

Very Short Answer Type

- When does the sudden and noticeable changes begin to appear in humans?
- What is puberty?
- Do both boys and girls grow at the same rate?
- Why the adolescent girls are at a higher risk of developing blood related health issues?
- What is the full form of WHO?
- What do you understand by balanced diet?
- Name the male hormone released by the testes.
- Name the female hormone released by the ovaries.
- What is the function of sweat glands?
- What is menopause?
- Testosterone and estrogen are secreted by which organs?
- What do you understand by the term substance abuse?
- Define menstrual cycle.
- What are the constituents of a balanced meal?
- What are hormones?

Short Answer Type

- Who was Dorothy Hodgkin?
- What are the changes in shape of the body of girls and boys during adolescence?
- Explain the development of sex organs in males and females.
- What are the various food sources that help in bone growth?
- Explain the changes occurring in the voice of boys and girls during puberty.
- Why sanitary pads are preferred over cotton cloth?
- What is the cause of the changes in human body during puberty?
- Write short note on Adam's apple.
- Explain about the secondary sexual characters.
- Enlist the changes in body structure, weight gain, and strength of boys and girls during adolescence.

Long Answer Type

- Describe various mental, intellectual and emotional changes taking place during adolescence.
- Elaborate on the reproductive cycle occurring in females.
- What are the factors which help to keep a good health during adolescence?
- What are the essential nutrients required for teenagers?
- How can bad habits like drug addiction harm adolescents?

Case Based Questions

Case I : 'X' is the period in life at which sexually immature boy or girl becomes mature and capable of reproduction. It happens to be at the age of 9 to 13 years in

girls and 10 to 14 years in boys. However, the period may vary from person to person. 'X' ends when an adolescent reaches the reproductive maturity.

CFQs

- What is represented by 'X' in the given passage?
(a) Puberty (b) Infancy
(c) Adulthood (d) Childhood
- Choose the correct statement regarding the case.
(a) Period X results in internal and observable changes in individuals.
(b) Reproductive maturity is the beginning of period X.
(c) X happens to be earlier in girls as compared to boys.
(d) All of these
- Most remarkable feature of period X is
(a) sexual immaturity
(b) reproductive maturity
(c) growth of voice box
(d) increase in height.
- What helps adolescents grow well during the period mentioned in given case?
(a) Eating nutritious food and exercising
(b) Avoiding school
(c) Watching TV all day
(d) Skipping meals
- At phase X, the testes produce which hormone?
(a) Estrogen (b) Insulin
(c) Progesterone (d) Testosterone

Case II : Reproductive health is a state of physical, mental and social well-being of a person in all matters relating to reproductive system at all stages of life. To maintain good reproductive health during adolescence, it is necessary to eat balanced diet, maintain personal hygiene, adequate physical exercise and to avoid taking any drug.

- How personal hygiene can be maintained by an adolescent?
 - By taking bath regularly to avoid bacterial infections due to increased secretion of sweat gland.
 - By changing sanitary pads after every 5-6 hours.
 - By regularly doing physical exercise to boost mood and mind.
 - Both (a) and (b).
- Adolescents should take a balanced diet to meet their nutritional requirement. Identify the mismatched pair.
 - Pulses – Growth of body
 - Fruits – Helps fight infection
 - Leafy vegetables – Make blood in body
 - Bread and *chapati* – Necessary for healthy bones
- How can adolescents avoid drug misuse?
 - by staying alone
 - by saying “yes” to friends
 - by saying “no” to drugs and making good choices
 - by avoiding all medicines
- Which of the following statements is incorrect for good reproductive health?
 - Balanced diet is important during stage of rapid growth and development.
 - Lack of cleanliness during teenage may lead to various RTIs.
 - Sexual transmitted diseases do not spread through transfusion of infected blood.
 - All of these
- What is the full form of RTI?
 - Reproductive Tract Insemination
 - Reproductive Tract Infertility
 - Reproductive Tract Infection
 - None of these



Life Processes in Animals

= *notes*
= *imp. stuff*

Learning Outcomes...

💡 Nutrition in Animals

💡 Human Digestive System

💡 Digestion in Ruminants

💡 Digestion in Birds

💡 Respiration in Animals

➤ Respiration in Humans

➤ Respiration through Gills



Introduction

All animals perform certain essential life processes necessary for their survival, such as nutrition, respiration, excretion and reproduction. These together are known as life processes.

Being heterotrophic, animals cannot make their own food and therefore depend on other organisms for their nutritional needs. The process of obtaining nutrition involves three main stages—intake of food, digestion of food and assimilation of it in the body.

In animals, including humans, **food is essential for survival. Food gives us the energy to move, grow and carry out all the activities that keep us alive.** The food we eat contains important substances like **carbohydrates, proteins and fats**. But our body cannot use these substances in their complex form. They need to be broken down into simpler forms. This process is called digestion.

This breaking down of food occurs inside our body in a long tube called the alimentary canal. The process starts in the **mouth** when we eat and ends at the **anus**. As the food travels through the alimentary canal, various organs and digestive juices work together to digest the food and absorb nutrients.

DO YOU KNOW?

- » *Paramecium*, a slipper-shaped, unicellular, microscopic organism is found in freshwater ponds, pools, streams, etc. In *Paramecium*, the constant movement of hair-like ciliary processes sweep food particles towards the mouth of the cell for ingestion.
- » *Hydra* is a simple multicellular organism, which possesses a number of **tentacles** around its mouth. The stinging cells present on the tentacles kill the actively moving prey. The food is then pushed into mouth through tentacles. Inside the body cavity, the digestive juices secreted by the surrounding cells help to digest the food.

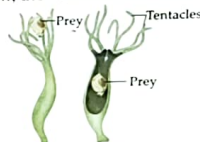


Fig.: Hydra catching prey

Starfish

- Starfish are marine invertebrates. They typically have a central disc and five arms, though some species have more than five arms.
- The mouth opens through a short oesophagus into a stomach. The stomach is divided by a constriction into a larger stomach and a smaller stomach. The starfish, pops out its larger stomach to engulf and digest food. When the starfish feeds on animals covered by hard shells of calcium carbonate, it opens the shell



COMPETITION WINDOW

- of the prey animal and then pops out its stomach through its mouth to engulf the soft animal inside the shell and releases enzymes to digest it.
- The stomach and the partially digested prey then goes back into the body of starfish and the food is slowly digested. Due to their ability to digest food outside the body, starfish can hunt prey much larger than their mouths. Their diets include clams and oysters, water insects, small fishes, shelled animals, etc. Some starfishes are not pure carnivores, supplementing their diets with algae or organic detritus.

Nutrition in Animals

Different animals eat different variety of food depending on what their bodies need and how they are built. Like bees and sunbirds feed on the sweet nectar of flowers, while infants and many animals feed upon their mother's milk. Some fishes and other aquatic animals e.g., sponges, many fishes, etc., filter tiny food particles from water, while snakes like pythons eat entire animals in a single swallow.

The food obtained by different types of organisms are complex in nature and needs to be broken down into simple soluble form by the process known as digestion. Digestion helps the nutrients present in food to get easily absorbed and used by the cells of organisms.

Human Digestive System

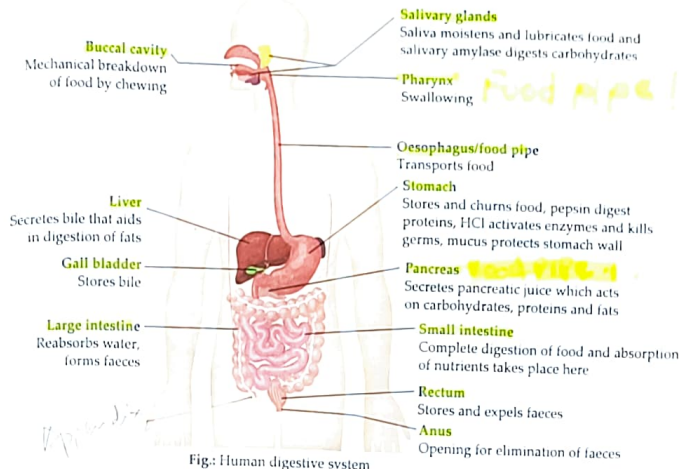
The digestive system of human beings consists of two main parts : **alimentary canal** (gut or digestive tract) and **digestive glands**.

The **alimentary canal** is a long tube which starts from the mouth and ends at the anus. It can be divided into various compartments :

- | | | |
|----------------------|------------------------------|---------------|
| (i) Buccal cavity | (ii) Food pipe or oesophagus | (iii) Stomach |
| (iv) Small intestine | (v) Large intestine | (vi) Anus |

Digestive glands such as **salivary glands** (in mouth), **gastric glands** (in stomach), **liver** and **pancreas** secrete digestive juices to convert complex substances of food into simpler ones. The inner wall of small intestine also secretes digestive juices.

The food components gradually get digested as food travels through the various compartments of alimentary canal.



The process of digestion in human beings involves both **mechanical and chemical actions**. **Mechanical digestion** is the physical breakdown of food by chewing and churning of food. In **chemical digestion**, chemical compounds, called **enzymes**, secreted by different organs and glands of the digestive tract break down large and complex substances like carbohydrates, fats and proteins into their simplest forms. Carbohydrates are converted into simple sugars (glucose); fats into fatty acids and **glycerol**; and proteins are converted into amino acids. The time taken for the food to go from mouth to the anus varies from person to person but can range from 12 to 48 hours, depending on the type of food eaten.

Let us understand the role of different parts of digestive tract during the journey of food through the digestive tract.

Mouth and Buccal Cavity

In human beings, food is taken in or **ingested** through mouth opening. The inner portion of mouth is called **buccal cavity**. Buccal cavity contains various structures such as teeth and tongue. The process of digestion starts in the **buccal cavity**, the moment we chew the food. The process of **chewing food** which involves movement of jaws and teeth is called **mastication**. It breaks down food into smaller pieces and allows the mixing of saliva with it. This is called the **mechanical digestion** of food. Chemical digestion of food in mouth is done by saliva. Saliva is a digestive juice, secreted by three pairs of salivary glands present in the mouth. **Saliva moistens the food and makes the chewed food slippery**. This helps in swallowing the food easily. Saliva contains an enzyme called **salivary amylase**. It helps in the breakdown of starch (complex carbohydrate) of the food into simple sugars that are easier to digest.

Saliva (secreted by salivary glands) is slightly acidic with a pH of 6.8. About 1000-1500 mL (1 to 1.5 L) of saliva is secreted per day in a normal human being.

ACTIVITY CORNER

To show that the saliva breaks down starch into sugars

► **Materials required :** Boiled rice, iodine solution, test tubes, dropper, chewed boiled rice (mixed with saliva).

► **Procedure :** (i) Take two test tubes and label them as A and B.
(ii) In test tube A, put one teaspoonful of boiled rice.

(iii) In test tube B, put one teaspoonful of boiled rice after chewing it for 3-5 minutes (mixing of saliva with boiled rice).

(iv) Add 3-4 mL of water in both the test tubes.

(v) Now add 2-3 drops of iodine solution in both the test tubes and observe the change in colour.

► **Observation :** When iodine is added to boiled rice in test tube A, it turns blue-black due to the presence of starch in rice. In test tube B, there is no change in the colour as the starch present in the rice has been acted upon by saliva (salivary amylase enzyme) and changed into simple sugars.

► **Conclusion :** Saliva breaks down starch into sugars.

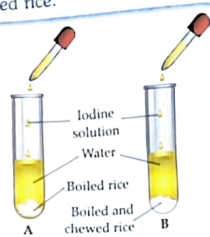


Fig.: Starch test (A) Boiled rice (B) Boiled and chewed rice

Teeth

An adult human being has 32 teeth, 16 in each jaw. Teeth break down food into smaller pieces by crushing and chewing. In human beings, teeth are of four different types based on their shapes and functions, i.e., incisors, canines, premolars and molars.

DO YOU KNOW?

► We should take care of mouth as it contains millions of bacteria. We should brush our teeth atleast twice in a day. It helps in removing the food particles that got stuck between the spaces of teeth. After every meal we should rinse our mouth thoroughly with water. It helps to maintain oral hygiene.

► **Incisors :** Incisors are the front teeth. These are chisel-shaped and are used for **biting and cutting**.

► **Canines :** On either side of the incisors, are present sharp, pointed teeth called canines. These are used for **piercing and tearing** the pieces of food such as meat.

► **Premolars and molars :** These teeth are present behind the canines. These are broad and strong with almost flat surface. Molars are larger than premolars.

COMPETITION WINDOW

Both premolars and molars are used for **crushing and grinding** the food.

► There are 4 incisors, 2 canines, 4 premolars and 6 molars in each jaw or we can say that there are 8 incisors, 4 canines, 8 premolars and 12 molars in the mouth of an adult human. Each tooth is rooted in a separate socket in the gums. The white substance that covers our teeth is called **enamel**. Enamel is the **hardest substance of human body**.

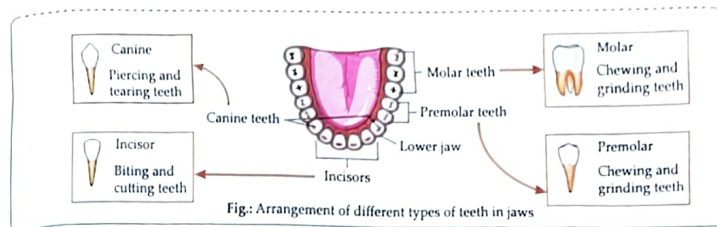


Fig.: Arrangement of different types of teeth in jaws

Tongue

The tongue is a fleshy muscular organ attached at the back to the floor of the buccal cavity. It is free at the front and can be moved in all directions.

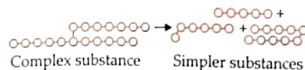
Tongue performs the following functions :

- The tongue helps us to taste the food.
- It moves food in our mouth between the teeth for chewing.
- It mixes saliva with food while chewing.
- It helps us to swallow the food.

Illustrations

1 Why is the food needed to be digested?

Ans.: The components of food such as carbohydrates, proteins, etc., are complex substances. These complex substances cannot be utilised as such. So, they need to be broken down into simpler substances to be used by cells and tissues of body. The breakdown of complex components of food into simpler substances is called digestion.



2 What is the mode of feeding in bees, butterflies and humming birds?

Ans.: Bees, butterflies and humming birds have sucking type of mode of feeding as they suck the nectar from flowers.

3 Which enzyme is responsible for the digestion of carbohydrates in buccal cavity in human beings?

Ans.: Salivary amylase

4 How does the mechanical and chemical digestion of food takes place in buccal cavity?

Ans.: In buccal cavity, mechanical digestion of food involves breaking it into pieces with the help of teeth and chemical digestion involves action of salivary amylase to convert food into simple form.

Oesophagus (or Food Pipe)

The tongue pushes the food from the mouth towards the pharynx. Pharynx is a common passage for food and air. The moist ball of food called bolus when swallowed reaches the food pipe or oesophagus. The human oesophagus is about 25 cm long muscular tube. It runs along the neck and the chest. Its function is to transfer the food from pharynx to the stomach. The muscles of oesophagus gently push food down to the stomach in a wave-like action (contractions and

relaxations) called **peristalsis**. Actually, this movement takes place throughout the alimentary canal to push the food forward.

Direction of the peristaltic movements is reversed during vomiting, which may be due to some factors such as taking stale food. In this situation food instead of moving downwards is pushed upwards leading to vomiting.

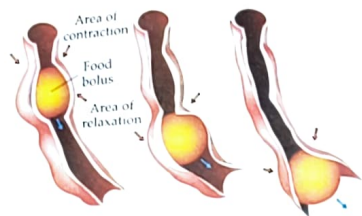


Fig.: Movement of food in oesophagus through peristalsis

DO YOU KNOW?

- It is always advised not to talk or laugh while eating and never to eat in a hurry. It is because by doing so, you may have **cough**, get **hiccups** or a **choking** sensation. This happens when food particles enter the windpipe. Pharynx is the common passage for food and air. It leads to food pipe and windpipe which run adjacent to each other. There is a flap-like structure, called **epiglottis**, which closes the opening of the windpipe while eating. This prevents the entry of food into the windpipe. If we talk while eating or eat in hurry, the epiglottis may not close and entry of food particles in the windpipe may result in choking, coughing, or hiccups.
- Sometimes we hear the gurgling or growling sounds from our stomach. It occurs when the mechanical digestion of food is taking place in the stomach by churning.

Stomach

The stomach is a J-shaped, thick-walled bag-like structure. It is the widest organ of alimentary canal. It receives food from food pipe at one end and opens into small intestine at the other end. The inner lining of stomach secretes digestive juice (or gastric juice) which contains mucus, hydrochloric acid and digestive enzymes.

Functions of stomach

- The thick muscular walls of stomach contract to churn the food and mix it with digestive juice produced by the stomach.
- Mucus protects the lining of stomach from the action of digestive enzymes and acids.
- Hydrochloric acid (HCl) kills bacteria that enter stomach along with food and it makes the medium in the stomach acidic to help the digestive enzymes to act.
- Digestive juice of stomach contains digestive enzymes, e.g., pepsin which cause the breakdown of proteins into simple substances (i.e., amino acids).
- Partial digestion of food occurs in the stomach.
- Food is stored in the stomach for few minutes to few hours depending on the type of food eaten. When food leaves stomach, it is like a thin paste, which is then moved down to small intestine where most of the digestion takes place.

Alexis St. Martin was a Canadian voyager who is known for his part in experiments on digestion in humans, conducted by an American army surgeon **William Beaumont** between 1822 and 1833. Alexis St. Martin in 1822, was badly shot with a musket in an accident. It seriously damaged the chest wall and made a hole in his stomach. Doctor Beaumont treated the wound and saved the life of patient. However, the hole (called **fistula**) in the stomach could never be closed and healed completely. Beaumont could literally watch the process of digestion by dangling food on a string into St. Martin's stomach, then later pulling it out to observe to what extent it had been digested. Alexis St. Martin allowed

COMPETITION WINDOW

the experiments to be conducted, because Beaumont had the illiterate St. Martin sign a contract to work as a servant. Beaumont found that the stomach was churning food. Its wall secreted a fluid which could digest the food. Dr. Beaumont also extracted a sample of gastric juices from Martin's stomach for analysis. He used it to digest bits of food in cups. This led to the important discovery that the stomach juices digest the food into nutrients the body can use; in other words, digestion was primarily a chemical process and not a mechanical one. He also observed that the end of the stomach opens into the intestine only after the digestion of food inside the stomach is completed.

Liver, Gall bladder and Pancreas

The liver is a reddish brown gland situated in the upper part of the abdomen on the right side. It is the **largest gland** in the body. It performs many important functions. The liver secretes **bile juice** which is stored in an elongated sac like structure called **gall bladder**.

Bile is not an enzyme, it is mildly basic in nature. It does not digest the food chemically. It plays an important role in the digestion of **fats**. The bile breaks up fats into tiny droplets that can be digested and absorbed more easily. The digestive juices then act on these tiny droplets to form simpler compounds known as fatty acids and glycerol. Bile also helps in neutralisation of acidic semi-solid food that comes from stomach.

The pancreas is a long, narrow, pale-pink coloured gland situated behind the stomach. The pancreatic juice acts on carbohydrates, fats and proteins and changes them into simpler forms, i.e., glucose, fatty acids + glycerol and amino acids respectively.

Small Intestine

Small intestine is the **longest portion of alimentary canal**. It is over 7.5 metres long and highly coiled. The digestion of food is completed in the small intestine. It receives bile juice from liver, pancreatic juice from pancreas and also secretes several digestive juices (intestinal juices) of its own. Thus, when the partially digested food from stomach reaches the small intestine, it is mixed with bile, pancreatic juice and intestinal juice; and the complete digestion of all the components of food takes place.

Functions of small intestine

- Proteins are broken down to amino acids.
- Carbohydrates are broken down to simple sugars, such as glucose.



Fig.: Human liver, pancreas and gall bladder

- (iii) Fats are hydrolysed into fatty acids and glycerol.
(iv) Absorption of digested food takes place in the intestinal wall.

DO YOU KNOW?

» Small intestine performs a variety of functions. Partially digested food coming from stomach is continuously mixed with intestinal juices and is completely digested and absorbed. About 90% absorption of all nutrients occurs in small intestine, the other 10% occurs in stomach and large intestine. Villi present all along the length of small intestine, significantly increase the surface area. Increased length and increased surface area of small intestine makes it efficient to carry out various functions.

Absorption of digested food in small intestine

The digested food can be utilised by our body only if it gets absorbed. The digested food can pass into the blood vessels in the wall of the intestine. This process is called **absorption**. Absorption of food occurs through thousands of finger-like outgrowths in the inner walls of the small intestine known as **villi** (singular = villus). The villi increase the surface area for absorption of the digested food. Each villus has a network of thin and small blood vessels close to its surface. A minute blind-ended lymph vessel called **lacteal** also occurs in each villus of the small intestine. Digested fats are absorbed into the lacteals. The food absorbed on the surface of villus passes into the blood in the capillaries.

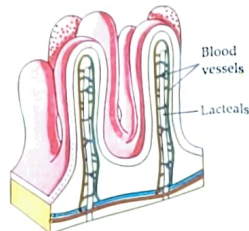


Fig: Villi with blood vessels in small intestine



Fig: Millet

Celiac disease is a condition in which the body's immune system reacts to gluten, a protein present in wheat, barley and rye. This reaction harms the inner lining of the small intestine, where nutrients are absorbed, leading to improper functioning of the intestine. The only effective way to manage celiac disease is by avoiding foods that contain gluten. Millets such as *jowar*, *bajra* and *ragi* serve as healthy alternatives since they are naturally free from gluten.

Assimilation

The absorbed substances are transported via the blood vessels to different organs of the body where they are used to build substances for growth, generate energy and repair worn out body cells. This is called **assimilation**. The food that remains undigested and unabsorbed enters into large intestine.

The fate of absorbed nutrients

- Carbohydrates**: During digestion, carbohydrates are broken down to glucose, which is used as a source of energy by the body. Glucose is oxidized to release energy inside the cells. Excess of glucose is stored in the cells of liver.
- Proteins**: During digestion, proteins are broken down to amino acids. The amino acids are used to build body tissues, including muscle and skin.
- Fats**: Fats are broken down into fatty acids and glycerol during digestion. Fats are also stored in the storage depots in various parts of body.

Large Intestine

The large intestine is wider and shorter than small intestine. It is about 1.5 metres in length. Large intestine is called large because its diameter is wider than small intestine. Its function is to absorb water and some salts from the undigested food material. Here, most of the water present in the undigested food material is absorbed. The waste food which is now semi-solid is stored in the last part of the large intestine called the **rectum**. It is then passed out of the body through the anus in the form of faeces. This is called **egestion**.

DO YOU KNOW?

- » The large intestine of human body houses about 100 trillion bacteria.

The relationship that exists between the body and its resident bacterial population is the **symbiotic relationship** by which both the partners are benefitted. In this relationship, these useful bacteria exert a positive influence on health. In return, the bacteria are provided with steady growth conditions and continuous supply of nutrients from human body.

Fermented foods like curd, buttermilk, *shrikhand*, *kanji*, pickles, *gundruk*, and *poita bhat* are good for a healthy digestive system and overall well-being. An ancient Ayurvedic text *Charaka Samhita*, highlights the role of easily digestible foods and the judicious use of spices like ginger, black pepper, and cumin to enhance digestion. The key factor maintaining digestive system emphasise on eating meals at proper timings, practising mindful eating, and avoiding overeating.

COMPETITION WINDOW

- » In human digestive system, starch (complex carbohydrate) is first broken down into **maltose** (simple carbohydrate) by amylase enzyme. Maltose is then broken down into **glucose** by another enzyme called maltase. Maltose is made up of two molecules of glucose.



- » The circulatory and respiratory systems of our body are directly linked to nutrition. The circulatory system transport the nutrients absorbed by the intestine to various parts of the body. The respiratory system provides oxygen to the cells for the oxidation of food to release energy.

- » **Constipation** refers to difficulty during defecation due to decreased motility of intestine. Faeces becomes dry and hard, and becomes difficult to pass. Constipation is a common problem which may be caused by habit of delaying defecation, taking low fibre diet, inadequate fluid intake, lack of physical exercise, emotional stress, side effects of some drugs, etc. The main treatment of constipation includes increased intake of water and fibre; and use of laxatives.
- » **Heartburn** is a discomfort or burning sensation in the chest and a bitter taste in throat or mouth. It is caused by the regurgitation (backward flowing) of acidic contents of stomach into the oesophagus.

Illustrations

5 What is the role of HCl secreted by inner lining of stomach wall?

Ans.: Hydrochloric acid (HCl) kills bacteria that enter stomach along with food and makes the medium of stomach acidic to help the digestive enzymes to act.

6 How does the oesophagus push the food down into stomach?

Ans.: Muscles in the wall of the oesophagus work in a coordinated manner by contractions and relaxations called peristalsis to push food and liquids down into the stomach.

7 Small intestine is over 7.5 metres long. How does such a long structure get accommodated in our body?

Ans.: The highly coiled nature of small intestine helps it to get accommodated in the small space of abdominal cavity in our body.

8 How does bile help in the digestion of fats?

Ans.: The bile breaks down fats into tiny droplets that can be digested more easily. The tiny droplets are acted upon by digestive juices to form simpler compounds.

9 What happens to food after it is digested

and absorbed by the blood?

Ans.: After the processes of digestion and absorption, assimilation of food takes place. The food which is absorbed into the blood, is transported to different parts of the body. It is used to provide energy and materials for growth and repair of body tissues.

10 Where do the digestive juices come from in small intestine?

Ans.: Some digestive juices are secreted by the cells of small intestine itself. Others come from liver (bile) and pancreas (pancreatic juice).

11 Mention the chief functions of large intestine in human digestive system.

Ans.: The chief functions of large intestine in human digestive system are as follows:

- Absorption of water and some mineral salts from the undigested food.
- Elimination of solid wastes as faeces.

12 Name the largest gland of our body.

Ans.: Liver

13 Define the term egestion.

Ans.: Egestion is the process of removal of undigested food (waste matter) from an organism's body.

Digestion in Ruminants

Ruminants are the hooved, plant eating (herbivorous) mammals which include cattle (cow, buffalo), goat, sheep, deer, yak, etc. Ruminants possess a unique digestive system that allows them to better use energy from fibrous plant material than other herbivores. Stomach of these animals consists of four chambers — rumen, reticulum, omasum and abomasum.

During grazing, they quickly swallow plenty of grass without much chewing. This is an adaptation by which they need very little time for feeding. A cow spends about 8 hours a day just chewing the food. The food is partially chewed and mixed with saliva. This partially chewed food is sent to rumen (large sac like structure). Here, partial digestion of food takes place as it harbours large population of anaerobic bacteria which help in fermentation of food. These microorganisms produce

enzyme (cellulase) which helps in digestion of cellulose (a complex carbohydrate present in the grass). This partly digested food called cud, is then pushed to the second chamber — reticulum. In fact, the rumen and reticulum are closely associated and their contents mix freely. When the animal is resting, the pulpy cud is brought back into the mouth in small quantities and the animal chews it further. This process is called rumination and thus, these animals are called ruminants (or cud-chewing mammals). After complete chewing, the food is sent to other parts of stomach i.e., abomasum and omasum; and ultimately to intestine for complete digestion and absorption of nutrients.

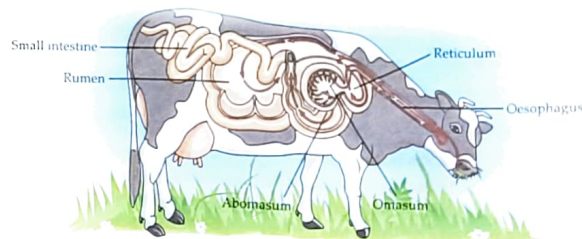


Fig.: Digestive system of a ruminant

Ruminants are able to digest cellulose as they contain bacteria and other microbes in their stomach which assist in the digestion of cellulose. Human beings cannot eat and digest cellulose as they do not have cellulose digesting bacteria in their stomach or intestine.

Digestion in Birds

Have you ever wondered how birds, like pigeons or parrots, get energy to fly high and sing cheerfully all day? Just like us, birds also need food for energy — but their digestive system works a little different.

Unlike humans, birds don't have teeth to chew their food. Instead, they use their beaks to pick and swallow food whole or in small pieces. Once swallowed, the food goes down the oesophagus and enters a special pouch called the crop, where it is stored and softened.

From the crop, the food moves into the stomach, which has two parts:

- Proventriculus** — this is the part where digestive juices are added to start breaking down the food.
- Gizzard** — a strong, muscular part that grinds the food, often with the help of small stones or grit that birds swallow.

After grinding, the food passes into the intestines, where nutrients are absorbed into the bird's body. Finally, the waste is expelled through the cloaca.

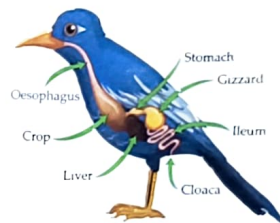


Fig.: Digestive system of a bird

Illustrations

14 Ruminants are able to digest cellulose present in grass whereas humans cannot. Why?

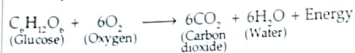
Ans.: Ruminants are able to digest cellulose as they contain bacteria in their stomach which assist in the digestion of cellulose. Human beings cannot eat and digest cellulose as they do not have cellulose digesting bacteria in their stomach or intestine.

Respiration in Animals

Different animals have different ways by which exchange of oxygen and carbon dioxide occurs in their bodies. Animals such as reptiles (snakes, lizards, crocodiles, etc.), birds and mammals (lion, cow, elephant, etc.) possess lungs like human beings. So, the process of respiration occurs through lungs in these animals. However, animals which do not have lungs, respire through other organs, e.g., fish respire through gills, frogs respire through skin (under water) and insects respire through spiracles, etc.

Aerobic Respiration

» When breakdown of glucose into carbon dioxide and water occurs in the body cells, in the presence of oxygen, it is called aerobic respiration. A large amount of energy is released during this process. Organisms that use oxygen for respiration are called **aerobes** (or aerobic organisms). Majority of animals and plants perform aerobic respiration.



Anaerobic Respiration

» When glucose is broken down without the use of oxygen, it is called anaerobic respiration. Organisms that can survive and

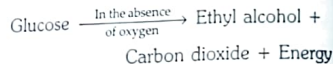
15 In what way does the basic structure of the stomach of a cow differ from that of a human?

Ans.: Stomach of cow is divided into four chambers—rumen, reticulum, omasum and abomasum. It is specialised for cud-chewing process. Whereas, stomach in a human is not separated into compartments. This basic difference in structure is mainly due to the difference in diets of ruminants and humans.

COMPETITION WINDOW

obtain energy in the absence of oxygen are called **anaerobes** (or anaerobic organisms). During anaerobic respiration, glucose is not completely broken down into carbon dioxide and water. It is broken down either into ethyl alcohol and carbon dioxide or into lactic acid. Very little amount of energy is released during anaerobic respiration. Anaerobic respiration occurs in yeast, some bacteria and parasitic flatworms.

» In **yeasts**, glucose breaks down into ethyl alcohol and carbon dioxide.



DO YOU KNOW?

Parasitic animals (e.g., roundworms and flatworms) that live in the alimentary canal of humans or other animals also respire anaerobically.

Respiration in Humans

The food that is digested by our body contains glucose, besides other nutrients such as amino acids, fatty acids, etc. These nutrients, after absorption are transported to each cell of the body. In the cells, oxidation of these nutrients takes place, e.g. glucose is oxidised (in the presence of oxygen) to release energy, water and carbon dioxide. This process is called respiration. It is also known as cellular respiration as it takes place in the cells. Cellular respiration takes place in the cells of almost all living organisms. It can be represented as:



Breathing is a part of respiration. Respiration is a **biochemical process** which involves breaking down of food to release energy. On the other hand, breathing is a mechanical process. It involves taking in of air rich in oxygen into the lungs and giving out of air rich in carbon dioxide from the lungs. The process of taking in of oxygen rich air into the lungs is called **inhalation** (or inspiration) and the process of giving out carbon dioxide rich air is known as **exhalation** (or expiration).

Inhalation and exhalation occur continuously and alternately. One breath consists of one inhalation and one exhalation. The number of times a person breathes in a minute is termed as **breathing rate** or **respiratory rate**. Human respiratory rate is measured when a person is at rest, and involves counting the number of breaths in one minute by counting how many times the chest rises.

DO YOU KNOW?

When breathing rate slows down to the extent that we do not get sufficient oxygen, we feel sleepy or drowsy.

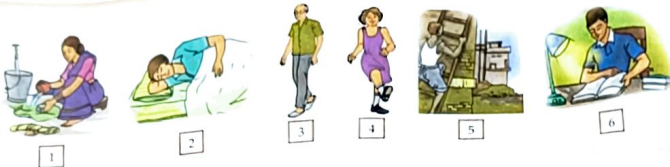
COMPETITION WINDOW

- » Normal breathing rate of a healthy adult person is 16-18 times/minute. Breathing rate may increase upto 25 times/minute during fast running or heavy exercise.
- » Breathing rate may increase during fever, illness or other medical conditions.
- » Infants and young children have higher breathing rate, as compared to adults.

» During brisk walking, fast running or exercising, we breathe fast and take deep breaths. This provides more oxygen and also more energy, as more food is oxidised. That is the reason we feel hungry after exercising, running or playing.

Illustrations

16 Given pictures (1-6) show the various activities carried out by people. Arrange these activities in the order of increasing rate of breathing.



Ans.: 2 → 6 → 3 → 1 → 5 → 4

17 Why do we feel hungry after a physical activity?

Ans.: During physical activities such as brisk walking, fast running, exercising etc. we need extra energy and so we breathe faster. This

supplies more oxygen to our cells and hence speeds up the process of breakdown of food (i.e., the process of respiration) and more energy is released. As during this process, more food is burnt (or oxidised), so we feel hungry after heavy physical activity.

Human Respiratory System

The human respiratory system consists of the nostrils, nasal cavity, pharynx, voice box (larynx), windpipe (trachea), bronchi and lungs. The lungs are situated in a space within the chest called the thoracic cavity or chest cavity. The walls of the thoracic cavity are supported by the ribs. The floor of the thoracic cavity consists of a dome shaped sheet of muscles called diaphragm.

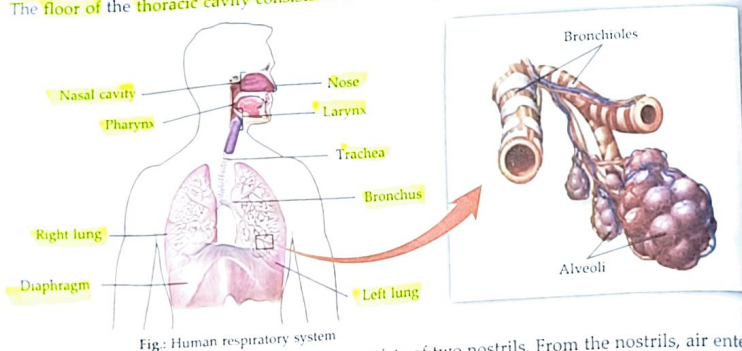


Fig.: Human respiratory system

We breathe in air through our nose which consists of two nostrils. From the nostrils, air enters into the nasal cavity.

The air around us has many impurities in it. However, our lungs require air which is moist, warm and clean. As the air we breathe in passes through the nasal cavity, it is moistened by

the slimy mucus secreted by the inner lining of the nose. The air also becomes warm by the blood circulating in the blood vessels of the nose. The mucus and the hair present inside the nose trap dirt, dust particles and disease-causing germs, and prevent them from entering the respiratory system. We do not get clean air if we breathe through our mouth instead of our nose. Therefore, we should always breathe through the nose as it will protect us from catching infections.

From the nasal cavity, the moist, warm and clean air passes through the pharynx to the windpipe or trachea. The upper part of the trachea contains larynx (or voice box) with the vocal cords which help in sound production. The windpipe branches into two smaller tubes called bronchi (singular-bronchus). Each bronchus enters into its respective lung.

Inside the lungs, the bronchi divide and redivide into bronchioles which further divide and finally end in air sacs or alveoli.

The air from trachea enters the bronchi, then the bronchioles followed by the alveoli. Thus, the air we breathe in eventually reaches these air sacs.

The air sacs are richly supplied with blood vessels (called alveolar capillaries). The oxygen present in the air we breathe in, enters into the blood contained in the alveolar capillaries.

The carbon dioxide present in the blood (as a waste product of respiration) passes out of the blood into the air sacs. Thus, exchange of gases takes place in the alveoli of the lungs.

The blood transports the oxygen to all the cells of the body, where it is used to get energy from food. The carbon dioxide released by the cells, goes back into the blood and is released into the air sacs of the lungs. It is then breathed out by us (exhalation).

DO YOU KNOW?

Smoking is a serious problem associated with society and extremely harmful to health. It damages the surface of alveoli and increases the risk of serious diseases, including lung cancer and other respiratory illnesses. Smoking releases toxic chemicals and harmful gases like carbon monoxide into the air. When healthy individuals inhale this polluted air, they experience passive smoking, which can be especially dangerous for children, pregnant women, and the elderly. Due to these risks, avoiding smoking helps protect both personal health and the well-being of those around us.

Mechanism of Breathing

Breathing involves the movements of the diaphragm and the rib cage.

Inhalation (or inspiration)

During inhalation, the ribs move upward and outward, and the diaphragm moves down (becoming flat). This increases the volume of thoracic cavity. The lungs, being elastic, also increase in size, and the air pressure in the lungs decreases. Air from the atmosphere having higher pressure rushes into the lungs through the respiratory passage.

Hence, the lungs get filled with fresh air.

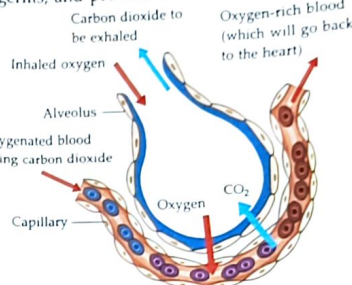


Fig.: Exchange of gases in alveoli

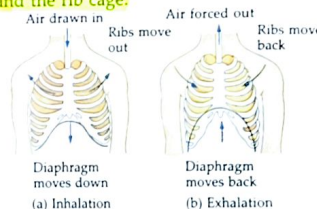


Fig.: Mechanism of breathing in human beings

Exhalation (or expiration)

During exhalation, the ribs move downward and inward, and the diaphragm moves up (again becoming dome shaped). When the ribs and diaphragm return to their original position, the volume of thoracic cavity decreases and so also of the lungs. This increases the air pressure inside the lungs and the air from the lungs is pushed out.

Breathing exercises are known to improve our respiratory health and cures us from many respiratory disorders. *Pranayama* is well known for improving respiratory health, mind relaxation, and concentration. It is practiced across different cultures in India as well as in different parts of the world. In Ladakh, people practise **Tummo breathing, a technique that improves lung function and helps keep the body warm even in cold weather. Some traditions combine deep breathing with chanting, using rhythmic breath control to enhance relaxation and mental clarity.**

Normally when we inhale, the unwanted particles (e.g. pollen grains, dust, etc.) get trapped in the hair present in the nasal cavity. However, sometimes these particles are not trapped in the hair and irritate the lining of nasal cavity. For example, during the COVID-19 pandemic, the SARS-CoV-2 virus affected the respiratory system, leading to breathing difficulties and often causing serious lung problems. If some foreign particles irritate the nasal cavity, we cough or sneeze and the particles come out of the body into air. As a result of sneezing, unwanted particles are thrown out and only clean air enters the lungs. When you sneeze, cover your nose so that the particles you sneeze out do not fall on others and are not inhaled by them.

- » During emergency conditions (e.g., fatal accidents, lung failure, etc.), when lungs do not function normally, artificial respiration is required. During artificial respiration,

COMPETITION WINDOW

oxygen gas is supplied to the affected person through an instrument called ventilator.

ACTIVITY CORNER

To demonstrate the mechanism of breathing

- » **Materials required :** A bell jar, two small balloons, a rubber sheet, a Y-shaped plastic or glass tube, a cork, etc.

Procedure :

- Fix two thin balloons to the two arms of the Y-shaped tube. Pass the Y-tube through one-holed rubber cork. Now, fix the cork along with the Y-tube and balloons in the mouth of the bell jar, as shown in the figure. Tie a rubber sheet tightly around the open rim of the base of the bell jar.
- Pull the rubber sheet down and observe the small balloons.
- Now, release the rubber sheet and allow it to flatten.
- Repeat the process three to four times.

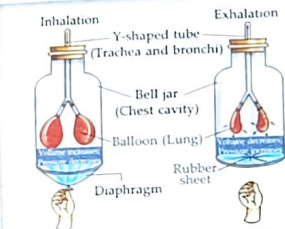


Fig.: To demonstrate the mechanism of breathing

» Observation :

- When the rubber sheet is pulled down, the small balloons expand.
- When the rubber sheet is released, the small balloons contract.

» Conclusion :

- The rubber sheet acts like the diaphragm in our respiratory system. When pulled down, it represents the downward movement of the diaphragm during inhalation. When released, it represents the relaxed state of the diaphragm during exhalation.

Composition of inhaled and exhaled air

We inhale and exhale air. But the composition of air inhaled and air exhaled is different. The inhaled air has more oxygen and less carbon dioxide, as compared to the exhaled air. This difference is given in the table below.

Table : Differences in the composition of inhaled and exhaled air

S.No.	Inhaled air composition		Exhaled air composition	
	(i)	Oxygen	Oxygen	16-17%
(ii)	Carbon dioxide	0.04%	Carbon dioxide	4-5%

DO YOU KNOW?

A pigment called **haemoglobin** present in red blood cells of the blood helps to transport oxygen. It combines with oxygen to form a temporary compound called **oxyhaemoglobin**. It is in this form that oxygen is carried through blood to different body cells. On reaching the cells, oxyhaemoglobin dissociates to release the trapped oxygen and the released oxygen diffuses into the cells. This oxygen is used for the oxidation (breakdown) of glucose to release energy. This is called internal respiration (or cellular respiration). Haemoglobin also combines with carbon dioxide and transports it from the cells to the lungs, where it can be breathed out.

Inspiration and Expiration : During inspiration, (or inhalation) the **external intercostal muscles** (which occur between the ribs) contract and pull the ribs and sternum upward and outward, thus increasing the volume of thoracic cavity. **Abdominal muscles** relax and allow compression of abdominal organs by the diaphragm. As a result, there is decrease of air pressure in the lungs.

The greater pressure outside the body causes air to flow rapidly into nostrils (external nares) and through the nasal cavity into internal nares which open into the pharynx. Thereafter the sequence of air flow is like this:
External nares → Nasal cavity → Internal nares

COMPETITION WINDOW

→ Pharynx → Glottis → Larynx → Trachea
→ Bronchi → Bronchioles → Alveolar ducts
→ Alveoli

During expiration, (or exhalation) the **internal intercostal muscles** contract so that they pull the ribs downward and inward, decreasing the size of the thoracic cavity. **Abdominal muscles** contract during expiration, and foul air goes outside from the alveoli in the following manner:

Alveoli → Alveolar ducts → Bronchioles → Bronchi → Trachea → Larynx → Glottis → Pharynx → Internal nares → Nasal cavity → External nares (nostrils) → Outside

ACTIVITY CORNER

To demonstrate that exhaled air contains more carbon dioxide as compared to inhaled air

► **Materials required :** Test tubes, lime water, two-holed stoppers, glass tubes, rubber tube, clips, etc.

► **Procedure :** Take two test tubes, each of them half-filled with lime water, using two-holed stoppers, glass tubes and clips, set up the apparatus as shown in figure. Use the rubber tube at the middle to breathe in and out through the mouth.

As you suck in air through the rubber tube, open clip X and close clip Y. The inhaled air passes through lime water in test tube A. As you exhale through the rubber tube, open clip Y and close clip X. The exhaled air passes through lime water in test tube B.

► **Observation and conclusion :** Lime water turns more milky in test tube B. In test tube A, lime water turns milky only upto a negligible amount. This shows that less amount of CO_2 is present in inhaled air whereas more amount of CO_2 is present in exhaled air.

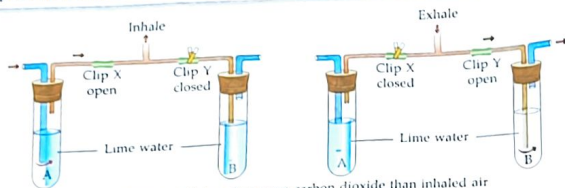


Fig.: Exhaled air has more carbon dioxide than inhaled air

Illustrations

18 Why is it always advised to breathe through nose and not through mouth?

Ans.: When we breathe in, the air passes through the nostrils and is moistened by the slimy mucus present in the nose. The mucus and the hair present inside our nose trap dust particles and disease-causing germs, and prevent them from entering the respiratory system. We do not get clean air if we breathe

through our mouth instead of nose. Therefore, we should always breathe through our nose. It will protect us from catching infections.

19 How do the movements of the diaphragm and the rib cage help us to exhale?

Ans.: During exhalation, the ribs move downward and inward, and the diaphragm moves up. Due to this, the volume of thoracic cavity decreases. This increases the air pressure inside the lungs and air from the lungs is pushed out.

Respiration through Gills

In fish, gills serve as the respiratory organs and help to obtain oxygen dissolved in water. Gills are the projections of skin made up of a large number of gill filaments, and are richly supplied with blood capillaries.

The carbon dioxide rich blood from body tissues is taken to gills for the exchange of respiratory gases. As water enters through the mouth of the fish, it flows over the gills. Oxygen dissolved in water diffuses into the blood capillaries. Carbon dioxide present in capillary blood diffuses into the water. The oxygenated blood from capillaries is taken to heart and pumped to different parts of the body.

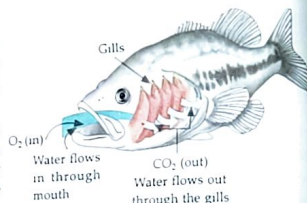


Fig.: Respiration through gills in fish

DO YOU KNOW?

► The skin of earthworm and leeches has numerous blood capillaries.

► Skin with a large number of capillaries is highly vascular.

► Whales and dolphins are marine mammals. Unlike fish, these mammals are not equipped with gills to extract oxygen from water. Instead, whales and dolphins need to come up to the surface of water to inhale and exhale air through their blowholes. Blowholes are the openings present at the top of their heads, which act as a passageway to the trachea, through which the air passes and reaches the lungs. When the dolphin and whale come up to the water surface for air, they open their blowholes, blow the foul air out of their lungs creating a plume of spray of air and water, and suck in fresh air. The oxygen they need to stay underwater for long period of time is stored in their blood and muscles.

Respiration in Frog

Respiration in frog can occur through skin, through lungs and through buccopharyngeal cavity. Respiration through skin occurs by simple diffusion method whereas respiration through lungs and through buccopharyngeal cavity occurs as follows:

Through lungs : Frogs have paired sac-like lungs. As in humans, the air enters the body of the frog through the nostrils, passes through the windpipe, and is received by the lungs. The mechanism of breathing, however, is different in frog from that in humans. However, the frog uses its throat to pump air into the lungs. The throat moves down drawing air into the mouth through the nostrils. The nostrils close and the throat moves up, forcing air into the lungs of

COMPETITION WINDOW

the frog. In humans, breathing is aided by the ribs, the diaphragm, and the chest muscles. However, the frog has no ribs or diaphragm, and its chest muscles are not involved in breathing.

Through buccopharyngeal cavity : Buccopharyngeal respiration in frog occurs when it is on land or partially immersed in water. There are present two external nares situated at the tip of the snout. Each external nares (nostril) leads into a nasal chamber which opens into the buccopharyngeal cavity by an internal nares. During this mode of respiration, the mouth and glottis are tightly closed while the nares are kept open.



EVERYDAY LIFE EXAMPLES

Life processes like nutrition, respiration, transportation, and excretion help animals to sustain their life. Nutrition refers to the process of intake of food such that organisms can fulfill their nutritional and energy-related needs. The nutrients need to be transported to different parts of the body through the process of transportation. The complex nutrients release their energy as they are broken down with the help of oxygen during cellular respiration. As many biochemical reactions take place in an organism, many waste and harmful products are also formed. These waste and harmful products need to be removed from the body by the process of excretion.

There are many examples in everyday life that are associated with life processes in animals like:

- ◆ **Saliva production:** When you smell or think of tasty food, your mouth starts watering. It is the saliva produced by salivary glands, which contains enzymes to start breaking down starch into sugar in oral cavity.
- ◆ **Breathing faster after exercise:** You may have noticed that a human or an animal breathes heavily after running or vigorous activity. This demonstrates the body's need for more oxygen to release energy from food through cellular respiration.
- ◆ **Fish using gills:** A fish in an aquarium use their gills to take in dissolved oxygen from the water is an example of external respiration in an aquatic environment.
- ◆ **Cows chewing cud:** Observing cows repeatedly bringing up partially digested food to chew it again shows rumination, a specialised form of digestion for fibrous plant material.

CONCEPT MAP

Life Processes in Animals

Nutrition

- Nutrition refers to the mode of intake of food to meet the energy requirements of body. The food is taken by different organisms using different modes of nutrition.
- The two major types of nutrition are autotrophic and heterotrophic nutrition.

Alimentary canal

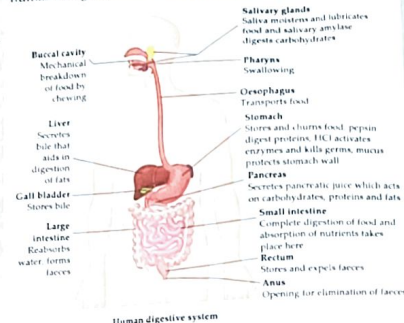
- Alimentary canal (or digestive tract) is a long tube which starts from the mouth and ends at the anus. It is subdivided into various compartments.
 - Mouth and Buccal cavity:** Food is taken into the body through mouth. It contains teeth and tongue. Different types of teeth help to chew the food and break it mechanically into small pieces. Saliva secreted by salivary glands converts starch into sugars. Tongue helps in mixing the food with saliva, swallowing, and tasting the food.
 - Oesophagus (food pipe):** It transfers the food from pharynx to stomach in a wave-like action called peristalsis.
 - Stomach:** Stomach is the J-shaped, thick-walled bag-like structure. It is the widest organ of alimentary canal.
 - Liver, Gall bladder and Pancreas:** Liver secretes bile juice which is stored in gall bladder. Pancreatic juice acts on carbohydrates, fats and proteins.
 - Small intestine:** It is the longest portion of alimentary canal. Complete digestion of all the food components takes place in small intestine.
 - Large intestine:** The large intestine is wider and shorter than small intestine. It helps to absorb water and some salts as well as helps to eliminate undigested food through anus.

Digestive glands

- The digestive glands make and store the digestive juices or digestive enzymes.
- The digestive glands include salivary glands, liver, pancreas, glands of stomach and small intestine.

Digestion in humans

The alimentary canal and the associated digestive glands together constitute human digestive system. The process of digestion in human beings involves both mechanical and chemical actions.



Digestion in other animals

- Animals adapt their alimentary canal structure based on dietary needs.
- **Ruminants** are the hooved, herbivorous mammals which include cattle, goat, sheep, deer, etc.
- Stomach of these animals consists of four chambers – rumen, reticulum, omasum and abomasum.
- They chew grass in mouth and swallow it into the stomach. They bring back partially digested food to the mouth for rumination.
- Some birds swallow tiny stones or grit or hard seeds that are broken down by extra chamber, gizzard present in their alimentary canal. It crushes and churns the food.

CONCEPT MAP

Life Processes in Animals

Respiration

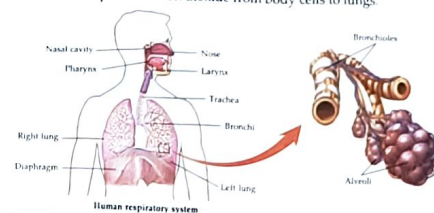
- The oxidation (or breakdown) of food in the cell with the release of carbon dioxide, water and energy is referred to as **respiration or cellular respiration**.
- The two main processes of respiration are – **external respiration** (involves inhalation and exhalation) and **internal respiration** (involves breakdown of food in the body cells, using oxygen, to release energy).

Respiration in other animals

- Different animals have different ways in which exchange of gases takes place in their bodies. e.g., respiratory organs are lungs in most reptiles, birds and mammals, gills in fish, spiracles and tracheae in insects, skin in earthworm, and skin, lungs and buccopharyngeal cavity in frog.

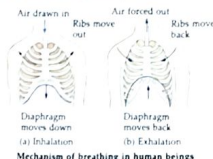
Human respiratory system

- Human respiratory system consists of the nostrils, nasal cavity, pharynx, larynx (voice box), trachea (windpipe), bronchi, and lungs.
- The air we breathe in eventually reaches the alveoli of lungs. Alveoli are surrounded by blood vessels. The oxygen present in the air goes into the blood present in blood vessels. The carbon dioxide present in blood passes out of the blood into the alveoli. Thus, **exchange of gases** takes place in the alveoli of the lungs.
- Haemoglobin pigment present in the red blood cells of the blood helps in the transport of oxygen from lungs to different body cells, and the transport of carbon dioxide from body cells to lungs.



Breathing

- Breathing is a mechanical (or physical) process which involves inhalation of oxygen into lungs and exhalation of carbon dioxide out of lungs. **One breath** consists of one inhalation and one exhalation.
- The number of times a person breathes in a minute is termed as **breathing rate (or respiratory rate)**.
- **Mechanism of breathing:**
 - During **inhalation**, the ribs move upwards and outwards and the diaphragm moves downwards. This increases the volume of thoracic cavity, decreases air pressure inside the lungs, resulting in entry of the air into the lungs through respiratory passage.
 - During **exhalation**, the ribs move downwards and inwards, and the diaphragm moves upwards. This decreases the volume of thoracic cavity, increases air pressure inside the lungs, resulting in pushing of air out of the lungs to the outside.



Solved Examples

1. Write short notes on the following:

- Mastication
- Functions of tongue
- Large intestine

Ans.: (i) **Mastication** : The process of chewing food which involves movement of jaws and teeth is called mastication. It breaks down the food into small particles which allows the mixing of saliva with it.

(ii) Functions of tongue is as follows:

- The tongue helps us to taste the food.
- It moves food in our mouth between the teeth for chewing.
- It mixes saliva with food while chewing.
- It helps us to swallow the food.

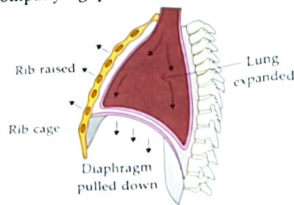
(iii) **Large intestine** : Large intestine is about 1.5 m in length. It is wider and shorter than the small intestine. Its functions are –

- to absorb water and some salts from the undigested food coming from small intestine.
- to eliminate waste matter in the form of faeces through anus.

2. Cows and buffaloes are usually seen chewing continuously even when they are not grazing. Explain.

Ans.: Cows and buffaloes are ruminants. These animals, during grazing, quickly swallow a large amount of food without much chewing. This partly chewed food is then sent to rumen (a part of stomach) where partial digestion of food takes place. This partially digested food is called cud and is brought back into the mouth in small lumps and animal chews it further. Thus, whenever we see a cow or buffalo sitting and chewing, at that time they are actually chewing the cud.

3. Study the given figure and answer the accompanying questions.



(i) Which breathing process has been shown in the given figure – inhalation or exhalation?

(ii) What changes take place during this process?

Ans.: (i) Inhalation has been shown in the given figure.
(ii) During inhalation, the ribs move upward and outward, and the diaphragm moves down. This increases the volume of thoracic cavity. The lungs, being elastic, also increase in size, and the air pressure in the lungs decreases. Air from atmosphere having higher pressure rushes into the lungs through the respiratory passage. Hence, the lungs get filled with fresh air.

4. How does the digestion of food takes place in stomach?

Ans.: In stomach, mechanical digestion of food occurs by alternate contraction and expansion of stomach wall. The thick muscular walls of stomach contract to churn the food and mix it with digestive juice, then chemical digestion of food starts during which digestive enzymes breakdown proteins into simpler forms (i.e., amino acids). Partial digestion of food takes place in stomach and this partly digested food then enters small intestine.

5. Why is the fibre-rich diet important for digestive system?

Ans.: Fibre-rich diet contains roughage (or fibre) which helps to stimulate waves of peristalsis along the alimentary canal especially in the region of small intestine. It results in pushing of food along the alimentary canal and facilitates removal of undigested waste matter from the body, thereby helping to reduce the chances of constipation.

6. Whenever we feel drowsy or sleepy, we start yawning. Does yawning help us in anyway?

(NCERT Exemplar)

Ans.: During drowsiness, our breathing rate slows down. The lungs do not get enough oxygen from the air resulting in yawning. Yawning brings extra oxygen into the lungs and removes more carbon dioxide and thus, helps us to keep awake.

7. What are the products of fat digestion?

Ans.: Products of fat digestion are fatty acids and glycerol.

8. The basic process of digestion and release of energy is the same in almost all animals. Justify this statement.

Ans.: Although modes of feeding are different in different animals, but almost all the animals secrete digestive juices to digest the food, i.e., to breakdown food into simple absorbable form. In the cells, absorbed food is then broken down into still simpler compounds with the release of energy through the process of respiration. Thus, the basic process of digestion and release of energy is the same in almost all animals.

9. Why small intestine is considered as the major site for digestion and absorption of nutrients?

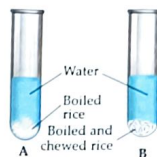
Ans.: Small intestine receives bile from liver, pancreatic juice from pancreas and releases intestinal juice of its own. Small intestine

completes the digestion of all the components of food. Carbohydrates are broken down into simple sugars such as glucose, fats into fatty acids and glycerol, and proteins into amino acids. About 90% of all absorption of nutrients occurs in small intestine. Thus, small intestine is the major site for digestion and absorption of nutrients.

10. How does gills help fishes in breathing?

Ans.: Fish use gills for exchange of gases. Gills are made up of large number of filaments, richly supplied with thin blood vessels called capillaries. As water enters through the mouth, it flows over the gills. The blood in the capillaries absorbs oxygen dissolved in water, and gives out carbon dioxide into the water.

11. Study the given experimental set-up. Test tube A contains boiled rice + 3-4 mL water; test tube B contains boiled and chewed rice + 3-4 mL water. What will you observe after pouring 2-3 drops of iodine solution in each test tube?



Ans.: After pouring iodine solution in test tube A it turns blue-black due to the presence of starch in boiled rice. In test tube B, there was no change in colour on adding iodine solution because starch has been converted into simple sugars during chewing by the action of enzyme salivary amylase.

12. What are villi? What role do they play in absorption of food?

Ans.: Villi are small finger-like projections present on the inner walls of small intestine. These increase the surface area for absorption of

digested food. Each villus has a network of thin and small blood vessels close to its surface. Surface of villi absorbs the digested food materials.

13. Name the animal which can breathe on land as well as in water.

Ans.: Frogs exhibit breathing both on land as well as in water. Inside water, they breathe through their skin. When on land, they breathe through their lungs or through buccopharyngeal cavity.

14. Breathing and respiration are not the same. Justify this statement.

Ans.: Respiration is a chemical process which involves breaking down of food to release energy.

Breathing is a mechanical process in which air rich in oxygen is taken into the body (inhalation) and air rich in carbon dioxide is given out (exhalation). Breathing is a part of respiration.

15. Refer to the given diagram which shows various parts of the human digestive system labelled as I, II, III, IV and V and answer the following questions.



(a) The labels I, II and III represent _____ and _____ respectively.

(b) Name the secretion produced by organ IV and explain how does it help in fat digestion.

(c) What is the function of organ V in human digestive system?

Ans.: (a) Oesophagus, stomach, pancreas.

(b) Organ IV is liver which produces bile. Bile is sent to small intestine where it gets mixed with fats present in food and converts them into small fat droplets. The latter are then digested and absorbed in small intestine.

(c) Organ V is gall bladder which stores the bile temporarily.

16. Pick the odd-one-out from each of the groups given below on the basis of respiratory organs. Give reason for your answer.

- (a) Cockroach, grasshopper, snail, ant
- (b) Lizard, cow, earthworm, snake
- (c) Crocodile, whale, dolphin, fish
- (d) Snake, tadpole, crow, goat

(NCERT Exemplar)

Ans.: (a) Snail is the odd one out. It is because cockroach, grasshopper and ant are all insects, which respire by means of tracheae, whereas most snails respire through gills.

(b) Earthworm is the odd one out because it respire through the skin by the process of diffusion, whereas lizard, cow and snake respire through lungs.

(c) Fish is the odd one out because fish respire through gills, whereas crocodile, whale and dolphin respire through lungs.

(d) Tadpole is the odd one out because it respire through gills, whereas snake, crow and goat respire through lungs.

17. How does exchange of gases take place in fish?

Ans.: Exchange of gases in fish takes place by gills. As water enters through the mouth of the fish, it flows over the gills, which are richly supplied with blood capillaries. Oxygen dissolved in water diffuses into the blood capillaries, while carbon dioxide present in capillary blood diffuses into water. The oxygenated blood from capillaries is taken to heart and pumped to different parts of the body while the carbon dioxide rich blood from body tissues is taken to gills for the exchange of gases.

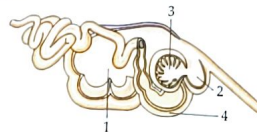
18. Name the different components, i.e. nutrients present in food.

Ans.: The different nutrients present in food are carbohydrates, fats, proteins, minerals, vitamins, etc.

19. How does food move in the opposite direction during vomiting?

Ans.: Food is pushed down by contraction and relaxation of the walls of the food pipe. This movement, called peristalsis takes place throughout the alimentary canal and pushes the food downwards. But when our stomach is completely full or after eating stale food, the food is not accepted by our stomach, therefore, it is pushed in the opposite direction and vomited out. Direction of peristaltic movements is reversed during vomiting.

20. Refer to the given diagram of a ruminant stomach and identify the parts labelled as 1, 2, 3 and 4.



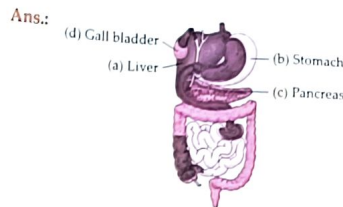
Ans.: 1-Rumen, 2-Reticulum, 3-Omasum, 4-Abomasum

21. Label the following parts in the figure given and name them.

- (a) The largest gland in our body.
- (b) The organ where protein digestion starts.
- (c) The organ that releases digestive juice into the small intestine.
- (d) The organ where bile juice gets stored.



(NCERT Exemplar)



22. You were blindfolded and asked to identify the drinks provided in two different glasses. You could identify drink A as lime juice and B as bitter gourd juice. How could you do it inspite of being blindfolded?

(NCERT Exemplar)

Ans.: We are able to identify different tastes with the help of different types of taste buds present in the tongue. Tongue is a fleshy muscular organ that bears a large number of taste buds that detect different tastes of food such as sweet, salty, bitter, sour, etc.

23. 'A' got her gall bladder removed surgically as she was diagnosed with stones in her gall bladder. After the surgery, she faced problems in digestion of certain food items when consumed in bulk. Can you tell which kind of food items would they be and why?

(NCERT Exemplar)

Ans.: The food items would be fats because bile juice which is stored by the gall bladder helps in the digestion of fats. Removal of gall bladder leads to difficulty in digestion of fats.

24. Trace the path of oxygen in humans during inhalation by completing the following blanks :

Nostrils → _____ → _____
→ _____ → Alveoli

Ans.: Nostrils → Nasal cavity → Pharynx → Trachea → Bronchi → Bronchioles → Alveoli

NCERT Section

1. Complete the journey of food through the alimentary canal by filling up the boxes with appropriate parts –

Food → Mouth → → Stomach →
→ → Anus

2. Sahil placed some pieces of *chapati* in test tube A. Neha placed chewed *chapati* in test tube B, and Santushti took boiled and mashed potato in test tube C. All of them added a few drops of iodine solution to their test tubes—A, B, and C, respectively. What would be their observations? Give reasons.

3. What is the role of the diaphragm in breathing?

- To filter the air
- To produce sound
- To help in inhalation and exhalation
- To absorb oxygen

4. Match the following.

Name of the part	Functions
(i) Nostrils	(a) Fresh air from outside enters
(ii) Nasal passages	(b) Exchange of gases occurs
(iii) Windpipe	(c) Protects lungs
(iv) Alveoli	(d) Tiny hair and mucus help to trap dust and dirt from the air we breathe
(v) Ribcage	(e) Air reaches our lungs through this part

5. Anil claims to his friend Sanvi that respiration and breathing are the same process. What question(s) can Sanvi ask him to make him understand that he is not correct?

6. Which of the following statements is correct and why?

Anu: We inhale air.

Shanu: We inhale oxygen.

Tanu: We inhale air rich in oxygen.

7. We often sneeze when we inhale a lot of dust-laden air. What can be possible explanations for this?

8. Paridhi and Anusha of Grade 7 started running for their morning workout. After they completed their running, they counted their breaths per minute. Anusha was breathing faster than Paridhi. Provide at least two possible explanations for why Anusha was breathing faster than Paridhi.

9. Yadu conducted an experiment to test his idea. He took two test tubes, A and B, and added a pinch of rice flour to the test tubes, half filled with water and stirred them properly. To test tube B, he added a few drops of saliva. He left the two test tubes for 35–45 min. After that, he added iodine solution into both the test tubes. Experimental results are as shown in figure. What do you think he wants to test?

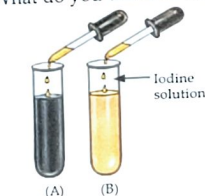


Fig. 2: Experimental results

10. Rakshita designed an experiment taking two clean test tubes, A and B and filled them with lime water as shown in the figure. In test tube A, the surrounding air that we inhale was passed on by sucking air from the pipe, and in test tube B, the exhaled air was blown through the pipe. What do you think she is trying to investigate? How can she confirm her findings?

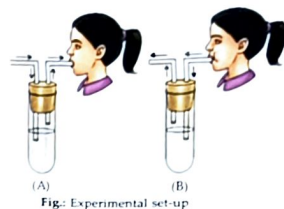


Fig. 3: Experimental set-up

NOTES

Exercise

Multiple Choice Questions

LEVEL - 1

Nutrition in Animals; Human Digestive System

- Read the following statements with reference to the villi of small intestine.
 - They have very thin walls.
 - They have a network of thin and small blood vessels close to the surface.
 - They have small pores through which food can easily pass.
 - They are finger-like projections.
 Identify those statements which enable the villi to absorb digested food.

- (i), (ii) and (iv)
- (ii), (iii) and (iv)
- (iii) and (iv)
- (i) and (iv)

(NCERT Page 122)

- The largest gland in human body is
 - pancreas
 - stomach
 - kidney
 - liver.
- Which secretion does not contain digestive enzymes?
 - Bile
 - Gastric juice
 - Intestinal juice
 - Pancreatic juice

(Page 122)

- _____ is the process of using the absorbed nutrients by the body cells for producing energy and for growth of the body.
 - Digestion
 - Absorption
 - Assimilation
 - Defecation

(Page 122)

- Which of these processes is not a part of nutrition?
 - Digestion
 - Absorption
 - Egestion
 - Locomotion

(Page 122)

- Which of the following set of organs is involved in the digestion of food?
 - Pancreas, liver and stomach
 - Stomach, oesophagus and kidney

- Heart, pancreas and stomach
- Liver, pancreas and lungs

(Page 122)

- The components of food are complex substances which cannot be utilised as such. So these are broken down into simpler substances. The process of breakdown of complex components of food into simpler substances is called
 - absorption
 - ingestion
 - digestion
 - assimilation.

(Page 122)

- The mechanical digestion of food starts in
 - oesophagus
 - mouth
 - small intestine
 - stomach.
- Which of the following statement is incorrect?
 - Saliva lubricates the food.
 - Tongue helps in tasting food.
 - Bile breaks down fat into small droplets.
 - Saliva helps to digest proteins.

(Page 124)

- Which part of the alimentary canal secretes acid?
 - Small intestine
 - Large intestine
 - Mouth
 - Stomach

(Page 124)

- The part of digestive system which helps in mixing food with saliva is
 - oesophagus
 - salivary glands
 - tongue
 - liver.

(Page 124)

- The liver and pancreas, release their secretions into
 - large intestine
 - small intestine
 - stomach
 - oesophagus.

(Page 128)

- Hydrochloric acid in stomach
 - helps in the digestion of fat
 - kills bacteria ingested with food
 - converts starch into sugars
 - converts proteins into amino acids.

(Page 128)

- Digestive juice of stomach contains pepsin enzyme which breaks down
 - complex carbohydrates into sugars
 - proteins into simpler form
 - fats into fatty acids
 - all of these.

(Page 125)

- The complete digestion and absorption of food occurs in
 - large intestine
 - small intestine
 - oesophagus
 - stomach.

(Page 126)

- Select the incorrect statement regarding human digestive system.
 - Digestive tract and associated glands together constitute the digestive system.
 - Tongue helps to mix the saliva with food and helps in swallowing.
 - Pancreatic juice acts only on carbohydrates and converts these into simpler form
 - Digestion of food is completed in small intestine.

(Page 126)

- Which of the following carries digested food to all parts of the body?
 - Blood
 - Mucus
 - Bile
 - Saliva

(Page 126)

- After digestion, the digested food is absorbed by blood while undigested food is stored temporarily in
 - rectum
 - stomach
 - small intestine
 - none of these.

(Page 127)

Digestion in Ruminants; Digestion in Birds; and Respiration in Animals

- What is the main function of the crop in birds?
 - To absorb nutrients
 - To add digestive juices
 - To grind the food
 - To store food and make it soft.

(Page 128)

- Which of the following organs are parts of the human respiratory system?
 - Pharynx
 - Nostrils
 - Gullet
 - Windpipe
 - Lungs

- II and V only
- II, IV and V only
- I, II, IV and V only
- All of these

(Page 128)

- Breathing is a process that
 - provides O_2 to the body
 - breaks down food to release energy
 - helps the body to get rid of CO_2
 - produces water in the cells.
 Which of the following gives the correct combination of functions of breathing?
 - (i) and (ii) only
 - (ii) and (iii) only
 - (i) and (iii) only
 - (ii) and (iv) only

(Page 128)

- Choose the correct order of terms that describe the process of nutrition in ruminants.
 - Swallowing $\rightarrow$ partial digestion $\rightarrow$ chewing of cud $\rightarrow$ complete digestion
 - Chewing of cud $\rightarrow$ swallowing $\rightarrow$ partial digestion $\rightarrow$ complete digestion
 - Chewing of cud $\rightarrow$ swallowing $\rightarrow$ mixing with digestive juices $\rightarrow$ digestion
 - Swallowing $\rightarrow$ chewing and mixing $\rightarrow$ partial digestion $\rightarrow$ complete digestion

(Page 128)

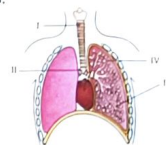
- In which part of a bird's digestive system food is broken down?
 - Gizzard
 - Crop
 - Cloaca
 - Proventriculus

(Page 128)

- Which of the following statements are correct about the bird's digestive system?
 - Birds have teeth to chew their food.
 - The gizzard helps in grinding the food.
 - The digested food goes down through the oesophagus.
 - Absence of small intestine.
 - Only 1 and 4
 - Only 2 and 3
 - Only 2, 3, and 4
 - All of the above

(Page 128)

- Given figure illustrates the internal structure of lungs.



The exchange of gases takes place at which labelled structure?

- I
- II
- III
- IV

(Page 129)

26. When a person breathes in, what happens to the diaphragm and to the ribs?

- Diaphragm becomes flattened, ribs move downwards and inwards
- Diaphragm becomes flattened, ribs move outwards and upwards
- Diaphragm becomes more curved, ribs move downwards and inwards
- Diaphragm becomes more curved, ribs move outwards and upwards

U

(Page 130)

27. The given figure illustrates the effect of exhaled air on lime water.

Lime water turns milky due to the presence of _____ in exhaled air.

- carbon monoxide
- carbon dioxide
- nitrogen
- oxygen

An

(Page 131)

28. Which of the following statements is correct?

- Inhaled air contains more oxygen than that of exhaled air.
- Exhaled air contains more oxygen than that of inhaled air.
- Inhaled air contains equal amount of carbon dioxide and oxygen.
- Exhaled air contains equal amount of carbon dioxide and oxygen.

U

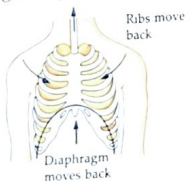
(Page 132)

29. Breathing is a _____ process and respiration is a _____ process.

- mechanical, mechanical
- physical, bio-chemical
- bio-chemical, physical
- chemical, chemical

(Page 132)

30. Given figure represents a breathing process.



During this process,
 (a) ribs move upward and outward
 (b) air from outside rushes into the lungs
 (c) air pressure inside the lungs increases
 (d) volume of thoracic cavity increases.

(Page 132)

31. Which of the following group of organisms respire through skin?

- Frog and fish
- Earthworm and frog
- Hydra and cockroach
- All of these

R

(Page 133)

LEVEL - 2

32. The given figure shows various parts of human digestive system labelled as I, II, III and IV. Identify the part that stores bile juice.

- I
- II
- III
- IV

An



33. Select the incorrect statement from the following statements.

- Large intestine is called large because its length is greater than small intestine.
- Minerals and vitamins do not need to be broken down during digestion.
- Mucus of stomach protects the lining of stomach from the action of enzymes and acids.
- Small intestine is smaller in diameter than large intestine.

U

34. The path followed by carbon dioxide to go out of the lungs is

- Alveoli → Bronchi → Bronchioles → Pharynx → Trachea → Nostrils
- Alveoli → Bronchioles → Bronchi → Trachea → Pharynx → Nostrils
- Nostrils → Pharynx → Trachea → Bronchioles → Bronchi → Lungs
- Nostrils → Trachea → Pharynx → Bronchi → Bronchioles → Lungs

An

35. Which of the following options shows the correct sequence of the flow of food from mouth to anus?

- Mouth → Oesophagus → Stomach → Small intestine → Large intestine → Anus
- Mouth → Large intestine → Oesophagus → Stomach → Small intestine → Anus
- Mouth → Small intestine → Large intestine → Oesophagus → Stomach → Anus
- Mouth → Stomach → Small intestine → Large intestine → Oesophagus → Anus

R

36. Which of the following correctly represents the approximate composition of inhaled air by humans?

- $O_2 = 21\%$ $CO_2 = 21\%$
- $CO_2 = 4.4\%$ $CO_2 = 0.04\%$
- $O_2 = 16.4\%$ $O_2 = 16.4\%$
- $CO_2 = 4.4\%$ $CO_2 = 0.03\%$

R

37. The digestion of carbohydrates starts in (i) with the help of enzyme (ii).

Which of the following options correctly fills the blanks and completes the given statement?

- (i) - buccal cavity; (ii) - pepsin
- (i) - stomach; (ii) - amylase
- (i) - buccal cavity; (ii) - amylase
- (i) - stomach; (ii) - pepsin

R

38. The given table illustrates the food components and their digested products.

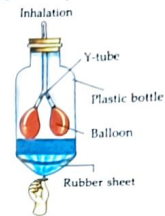
Food components	Digested products
P	Amino acids
Carbohydrates	Q

Select the correct option for P and Q.

- P - Fats; Q - glycerol
- P - Proteins; Q - glucose
- P - Fats; Q - glucose
- P - Proteins; Q - glycerol

An

39. Study the given experimental set-up.



Which of the following statements holds true when the rubber sheet is pulled down?

- Air pressure inside the bell jar decreases
- The two small balloons expand
- This is similar to the contraction of lungs during exhalation
- Both (a) and (b)

Ap

40. Match the respiratory organs in column I with the animals in which they are present in column II and select the correct option.

Column I	Column II
A. Gills	(i) Earthworm
B. Lungs	(ii) Fish
C. Skin	(iii) Crocodiles
D. Spiracles	(iv) Grasshopper

- A - (iv), B - (ii), C - (iii), D - (i)
- A - (ii), B - (i), C - (iii), D - (iv)
- A - (ii), B - (iii), C - (i), D - (iv)
- A - (iii), B - (ii), C - (i), D - (iv)

R

41. The given diagram shows parts of human alimentary canal labelled as I and II.



Identify I and II and select the correct statement regarding them.

- I represents small intestine which is the longest part of alimentary canal.
- II represents large intestine which is the longest part of alimentary canal.

- (c) II represents small intestine which is the widest part of alimentary canal.
(d) I represents large intestine which is the widest part of alimentary canal. **An**

42. In human respiratory system, the trachea is divided into two (i), which enter into two respective lungs. Inside the lungs, (i) divide and redivide into (ii) which further divide and finally end into (iii).
Select the correct option to complete the above paragraph.

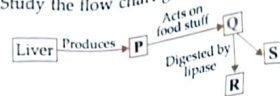
- | | | |
|-----------------|-------------|-------------|
| (i) | (ii) | (iii) |
| (a) bronchi | bronchioles | alveoli |
| (b) bronchioles | bronchi | alveoli |
| (c) bronchi | spiracles | bronchioles |
| (d) air sacs | bronchi | bronchioles |

43. Match column I with column II and select the correct option.

- | Column I | Column II |
|--|-------------------|
| A. Mechanical breakdown of food by chewing | (i) Stomach |
| B. Passage that connects mouth to stomach | (ii) Gall bladder |
| C. Chemical digestion of protein starts | (iii) Mouth |
| D. Stores bile | (iv) Oesophagus |
- (a) A - (iii), B - (iv), C - (i), D - (ii)
(b) A - (iii), B - (iv), C - (ii), D - (i)
(c) A - (iv), B - (iii), C - (i), D - (ii)
(d) A - (ii), B - (iii), C - (iv), D - (i)

44. Which of the following statements is incorrect?
(a) As a result of sneezing, unwanted particles are thrown out of the body.
(b) Trachea is divided into two bronchi which enter into respective lungs.
(c) The alveoli of lungs are not supplied with blood vessels.
(d) Breathing rate does not remain the same under all conditions. **U**

45. Study the flow chart given below.



Select the correct option for P, Q, R and S.

- | P | Q | R | S |
|---------------------|----------|-------------|-------------|
| (a) Digestive juice | Proteins | Fatty acids | Glycerol |
| (b) Bile | Fats | Fatty acids | Glycerol |
| (c) Bile | Proteins | Amino acids | Peptides |
| (d) Digestive juice | Fats | Glucose | Amino acids |

46. (i) digestion involves breakdown of large, (ii) food molecules into smaller soluble ones with the help of (iii).
Which of the following options fills the blanks correctly and completes the given statement?

- (a) (i)-Physical, (ii)-soluble, (iii)-enzymes
(b) (i)-Chemical, (ii)-insoluble, (iii)-enzymes
(c) (i)-Thermal, (ii)-insoluble, (iii)-heat
(d) (i)-Mechanical, (ii)-soluble, (iii)-heat **R**

47. Which of the following statements is/are correct regarding human respiratory system?
(a) Exchange of gases takes place in the alveoli which are surrounded by blood vessels.
(b) The blood takes oxygen to the cells where cellular respiration occurs.
(c) As the air we breathe in passes through the nostrils, it is moistened by the slimy mucus present in the nose.
(d) All of these

48. Select the correct statement out of the following.

- (a) Normally we breathe about 16-18 times per minute.

- (b) Breathing rate increases during physical activity.
(c) Breathing rate remains same under all conditions.
(d) Both (a) and (b) **U**

49. Match column I with column II and select the correct option.

- | Column I | Column II |
|---|----------------------|
| A. Maximum absorption of nutrients | (i) Anus |
| B. Absorption of water from undigested food | (ii) Small intestine |
| C. Opening through which waste matter is expelled out | (iii) Stomach |
| D. Secretion of HCl | (iv) Large intestine |
- (a) A - (ii), B - (iii), C - (i), D - (iv)
(b) A - (ii), B - (iv), C - (i), D - (iii)
(c) A - (iv), B - (ii), C - (i), D - (iii)
(d) A - (i), B - (iii), C - (ii), D - (iv) **R**

50. Given boxes contain characteristics of organ A and organ B of human body.

Organ A	Organ B
- Food is completely digested here.	- Water absorption takes place here.
- The substance produced is carried to all parts of body.	- The substance produced is expelled out of body.

Refer the boxes and select the option which correctly identifies organ A and B.

- | Organ A | Organ B |
|---------------------|-----------------|
| (a) Small intestine | Oesophagus |
| (b) Stomach | Large intestine |
| (c) Large intestine | Liver |
| (d) Small intestine | Large intestine |

51. Which of the following structures closes the wind pipe to prevent the entry of food into it during eating?

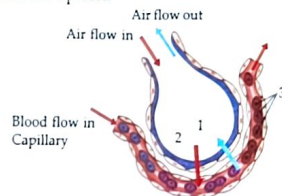
- (a) Pharynx
(b) Food pipe
(c) Epiglottis
(d) Tongue **U**

52. In humans, the process of digestion begins in the (I) where food is broken into small pieces by the teeth. The (II) in the mouth secrete saliva which contains an enzyme called (III). This enzyme changes (IV) to sugars.

Read the above paragraph and select the correct option for (I), (II), (III) and (IV).

- | (I) | (II) | (III) | (IV) |
|-------------------|-----------------|------------------|----------|
| (a) mouth | tongue | salivary amylase | starch |
| (b) buccal cavity | salivary glands | salivary amylase | starch |
| (c) mouth | salivary glands | pepsin | proteins |
| (d) stomach | gastric glands | pepsin | proteins |

53. Identify the labels (1, 2 and 3) in the given figure showing exchange of gases between alveoli and blood capillaries. Select the correct option.



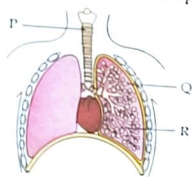
	1	2	3
(a)	CO ₂	O ₂	RBCs
(b)	O ₂	CO ₂	RBCs
(c)	RBCs	O ₂	CO ₂
(d)	CO ₂	RBCs	O ₂

54. The simple form of nutrients are obtained after the digested food gets absorbed in (I). The remaining matter goes into the (II) where excess of (III) is absorbed. The remaining waste is called (IV) and it is pushed into the rectum where it is stored before leaving the body through anus.

Read the above paragraph and select the correct option for (i), (ii), (iii) and (iv).

- (I) small intestine (II) rectum (III) water (IV) urine
(a) small intestine (b) stomach (c) large intestine (d) small intestine
- (i) small intestine (ii) rectum (iii) water (iv) bolus
(a) small intestine (b) stomach (c) large intestine (d) small intestine
- (i) small intestine (ii) rectum (iii) water (iv) faeces
(a) small intestine (b) stomach (c) large intestine (d) small intestine

55. Identify the labels P, Q and R in the given figure and select the correct option.



	P	Q	R
(a)	Trachea	Alveoli	Bronchiole
(b)	Bronchus	Alveoli	Lungs
(c)	Trachea	Bronchi	Alveoli
(d)	Bronchiole	Bronchi	Alveoli

Ans

56. During inhalation, the ribs move (i) and (ii), and the diaphragm moves (iii). This (iv) the volume of thoracic cavity and the air pressure inside the lungs (v). Complete the above paragraph by selecting the correct option.

- (a) (i) upward, (ii) outward, (iii) down, (iv) increases, (v) decreases
(b) (i) downward, (ii) inward, (iii) up, (iv) decreases, (v) increases
(c) (i) upward, (ii) outward, (iii) down, (iv) decreases, (v) increases
(d) (i) downward, (ii) inward, (iii) up, (iv) increases, (v) decreases

Ans

57. In human body, the absorption of water takes place in (i), while storage of faecal material takes place in (ii).

- (a) (i) - small intestine; (ii) - anus
(b) (i) - stomach; (ii) - rectum
(c) (i) - large intestine; (ii) - anus
(d) (i) - large intestine; (ii) - rectum

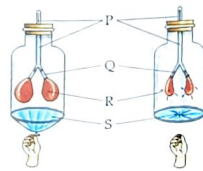
58. What happens when we breathe in?

- I. Size of our chest increases
II. Size of our chest decreases
III. Carbon dioxide is removed from our body
IV. Air enters our lungs
(a) I and III are correct.
(b) I and IV are correct.
(c) II and III are correct.
(d) II and IV are correct.

59. Which of the following options is correct regarding the percentage of oxygen and carbon dioxide in inhaled and exhaled air?

- (a) Inhaled air - 16.4% oxygen, 21% carbon dioxide
(b) Exhaled air - 0.04% oxygen, 16.4% carbon dioxide
(c) Inhaled air - 21% oxygen, 0.04% carbon dioxide
(d) Exhaled air - 0.04% oxygen, 21% carbon dioxide

60. In mammals the diaphragm is a muscular membrane that divides the thorax from abdomen. It plays an essential role in breathing. Which of the following labelled parts represents diaphragm in the given figure?



- (a) P (b) Q (c) R (d) S

LEVEL - 3 (HOTS)

Competency Focused Questions (CFQs)

61. Digestive juices were collected from three different regions of the alimentary canal. Drops of these juices were added to wells made in an agar of starch as shown below.

Well 1	Well 2	Well 3
Agar containing starch		

After an hour, the wells were rinsed with distilled water and flooded with iodine solution. The results are shown below.

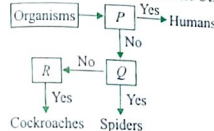
Well	1	2	3
Colour of iodine solution	Blue-black	Yellow - brown	Yellow - brown

Which of the following correctly identifies the regions of the alimentary canal from which the three digestive juices were obtained?

	Well 1	Well 2	Well 3
(a)	Mouth cavity	Small intestine	Stomach
(b)	Mouth cavity	Stomach	Small intestine
(c)	Small intestine	Mouth cavity	Stomach
(d)	Stomach	Mouth cavity	Small intestine

Ans

62. Refer to the given flow chart showing the characteristics P, Q and R which are present in some animals and absent in others.



Identify P, Q, R and select the correct option.

- P Q R
(a) Diaphragm Book lungs Operculum
(b) Throat Spiracles Book lungs
(c) Lungs Book lungs Spiracles
(d) Diaphragm Operculum Spiracles

63. Refer to the given diagram of human digestive system and select the correct option regarding its labelled parts P, Q, R, S and T.



- (a) R secretes digestive enzymes viz. amylase, lipase, pepsin, trypsin, etc.
(b) P secretes bile juice whereas T secretes hydrochloric acid.
(c) S is longer in cow as compared to tiger to aid the digestion of cellulose rich food.
(d) Q is the largest digestive gland which secretes different enzymes for digestion of fats, proteins and carbohydrates.

64. Which one of the following is a possibility for most of us in regard to breathing, by making a conscious effort?

- (a) The lungs can be made fully empty by forcefully breathing out all air from them.
(b) One can breathe out air through eustachian tube by closing both nose and mouth.
(c) One can consciously breathe in and breathe out by moving the diaphragm alone, without moving the ribs at all.
(d) One can breathe out air totally without oxygen.

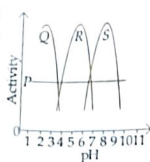
65. Read the given paragraph where few words have been italicised.

Buccal cavity of humans contains *two* pairs of salivary glands which secrete digestive juices that help in digestion of *proteins* present in food. From here food enters stomach where digestion of mainly *fats* takes place. As this semi-digested food enters small intestine complete digestion of food occurs. Small intestine receives bile juice from *pancreas* which *digests* fats. Most absorption of water occurs in *large* intestine.

- Select the correct option regarding this.
(a) *Two* should be replaced by *four* whereas *proteins* should not be replaced as it is correctly mentioned.

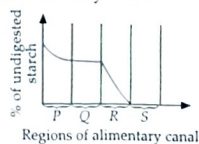
- (b) Fats should be replaced by carbohydrates and pancreas should be replaced by liver.
 (c) Digests should be replaced by emulsifies.
 (d) Large should be replaced by small. **An**

66. The given graph shows the effect of pH on the activity of four different enzymes (P, Q, R and S). Which of the following correctly identifies enzyme found in the stomach, small intestine and one not affected by pH?



	Stomach	Small intestine	Unaffected by pH
(a)	R	S	Q
(b)	S	P	R
(c)	P	R	Q
(d)	Q	S	P

67. Refer to the given graph representing percentage of undigested starch in different parts of alimentary canal.



Identify regions of alimentary canal P – S and select the correct statement.

- (a) P could be mouth, Q could be oesophagus, R could be stomach and S could be small intestine.
 (b) Digestion of food is completed in R which also possesses many finger-like projections in its wall that provide a large surface area for the absorption of food molecules.
 (c) Q secretes digestive juice which contains mucus, and enzyme amylase.
 (d) P makes the medium acidic to help the digestive enzyme to act upon food stuff.

68. Various technical terms related to respiratory system of different organisms are hidden in the given word grid.

R	M	S	L	U	N	G
I	G	I	L	L	N	T
B	A	O	I	D	O	R
C	R	M	Y	M	S	A
A	H	A	A	N	T	C
G	P	T	B	Z	R	H
E	A	A	T	S	I	E
P	I	R	A	C	L	A

Which of the following is not present in the given word grid?

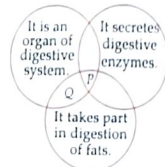
- (i) The air tubes of insects.
 (ii) Skeletal structure which protects the lungs.
 (iii) The opening through which humans inhale air.
 (iv) Small opening on the sides of the body of an insect.
 (v) A flap like structure which covers the glottis at the time of swallowing in humans.
 (vi) Respiratory organ of fish.
 (a) (i) and (ii) only
 (b) (ii), (v) and (vi) only
 (c) (iii) and (iv) only
 (d) (iv) and (v) only

69. Varun got his gall bladder removed surgically as he was diagnosed with stone in his gall bladder. Which of the following problems is Varun most likely to face after surgery?

- (a) Varun would not be able to digest carbohydrates and proteins properly because gall bladder stores bile which consists of amylase and pepsin enzymes.
 (b) Varun would not be able to digest fats properly because gall bladder stores bile juice which helps in emulsification of fats.
 (c) Varun would not be able to digest fats properly because gall bladder secretes bile juice which helps in digestion of fats.

- (d) Varun would not be able to absorb vitamin B and C properly because gall bladder secretes bile which helps in absorption of these vitamins. **Ap**

70. Refer to the given Venn diagram.



Identify P and Q and select the correct statement.

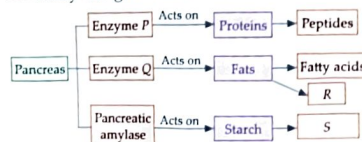
- (a) Q produces a juice which gets mixed with fats present in food and converts it into small fat droplets.
 (b) Q secretes a digestive juice which acts on carbohydrates, fats and proteins and converts them into glycerol, fatty acids and amino acids, respectively.
 (c) P could be the smallest organ of digestive system whereas Q could be the longest portion of alimentary canal.
 (d) P absorbs water and salt from undigested food whereas Q is the site of absorption of digested food. **An**

71. Swati took some boiled rice grains in test tube 'X' while Priya took boiled and chewed rice grains in test tube Y. Both of them poured 1 – 2 drops of iodine solution into the respective test tubes. What would they observe and why?

- (a) After pouring iodine solution in test tubes X and Y, colour of solutions of these tubes will change to blue black due to presence of starch.
 (b) After pouring iodine solution in test tube X, the colour of its solution will change to blue black due to presence of starch in boiled rice whereas colour of solution in test tube Y will not change because of digestion of starch into sugar by the action of salivary amylase during chewing.

- (c) After pouring iodine solution in test tube X, the colour of its solution will change to blue black due to the presence of sugar in boiled rice whereas colour of solution in test tube Y will not change because of digestion of starch into glycerol by the action of lipase enzyme during chewing.
 (d) After pouring iodine solution in test tube X, the colour of its solution will remain unchanged due to presence of starch in boiled rice whereas colour of solution of Y will change to blue black due to conversion of starch into fatty acids and glycerol by the action of pepsin enzyme during chewing.

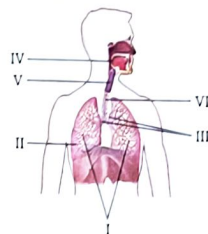
72. Study the given flow chart.



Select the option which correctly identifies P – S.

- P Q R S
 (a) Trypsin Amylase Glycerol Glucose
 (b) Pepsin Trypsin Glycerol Fructose
 (c) Trypsin Lactase Galactose Sucrose
 (d) Trypsin Lipase Glycerol Maltose **An**

73. Given below is the diagrammatic sectional view of the human respiratory system. Which set of three parts out of I-VI have been correctly identified?



- (a) (I) Bronchioles, (IV) Pharynx, (VI) Bronchus
 (b) (III) Bronchi, (IV) Pharynx, (VI) Trachea
 (c) (I) Bronchioles, (II) Alveolus, (VI) Bronchus
 (d) (II) Alveolus, (IV) Larynx, (V) Glottis

► Fill in the Blanks

- External respiration refers to taking in air rich in _____ and giving out air rich in _____.
- The process by which nutrients pass into the blood vessels in the wall of intestine is called _____.
- Bees, butterflies and humming birds _____ nectar from flowers.
- The _____ and _____ together constitute the human digestive system.
- Trachea divides into two branches called _____.
- Canines are the sharp, pointed teeth, used for _____ and _____.
- Under normal conditions, a person breathes _____ times a minute.
- Earthworms and leeches breathe through their moist and slimy _____.
- During digestion, fats are broken down into _____ and _____.
- _____ is the widest organ of an alimentary canal.
- Small intestine receives secretions from the _____ and _____ and also secretes its own intestinal juice.
- Saliva breaks down starch into _____.
- _____ are able to digest cellulose due to the presence of cellulose digesting bacteria in their stomach.
- During _____, the air pressure in the lungs decreases and fresh air enters into the lungs.
- _____ is a common passage for food and air.

- Muscles of the _____ and the _____ help in the breathing movements.
- Chest increases in size during _____ and decreases in size during _____.
- _____ is a sheet of muscle present below the lungs.
- About 90% of absorption of nutrients occurs in _____.
- During _____, the ribs move downward and inward, and the diaphragm moves up.
- Fish possess _____ for the exchange of gases through diffusion.

► True or False

- Villi are small projections in the inner wall of small intestine that helps in absorption of food.
- Inhaled air contains about 21% of oxygen and exhaled air contains about 16.4% of oxygen.
- Chemical digestion of food is completed in large intestine.
- Tongue helps in mixing of saliva with food.
- Inside the lungs, bronchi divide and redivide into bronchioles which finally end in alveoli.
- Air pressure in the lungs increases when lungs expand during inhalation.
- Bile juice is produced by gall bladder.
- Liver is the largest gland in human body.
- Digestion of proteins begins in buccal cavity.
- Whales and dolphins possess gills for respiration.
- Pancreatic juice acts only on carbohydrates.
- Cows, buffaloes and camels feed on grass very slowly during grazing.
- Mastication is the chewing of food to break it into smaller pieces.
- Breathing is the process of release of energy by the oxidation of food.

- Stomach does not produce any digestive enzymes.
- In amphibians such as adult frogs, exchange of gases takes place through gills.
- During exhalation, ribs move upwards and outwards whereas the diaphragm moves downwards.
- Breathing in humans takes place with the help of lungs.
- In humans, oxygen is absorbed and carbon dioxide is released during breathing.

► Assertion & Reason Type

Directions : In the following questions, a statement of assertion is followed by a statement of reason. Mark the correct choice as :

- (a) If both assertion and reason are true and reason is the correct explanation of assertion.
 (b) If both assertion and reason are true but reason is not the correct explanation of assertion.
 (c) If assertion is true but reason is false.
 (d) If assertion is false but reason is true.

- Assertion** : Oesophagus transfers food from mouth to the stomach.
Reason : Windpipe assists the oesophagus to transfer food from mouth to the stomach.
- Assertion** : During inhalation, the ribs move downwards and inwards while the diaphragm moves upwards.
Reason : During inhalation, volume of thoracic cavity increases.
- Assertion** : Food stays in stomach for a few minutes to few hours depending on the type of food eaten.
Reason : Digestion of starch begins in the stomach and digestion of proteins completes in the stomach.
- Assertion** : Bile is produced by liver and stored in gall bladder.
Reason : Bile breaks down fats into simpler substances, i.e., fatty acids and glycerol.

- Assertion** : The number of times a person breathes in a minute is termed as the breathing rate.

Reason : During breathing, inhalation and exhalation processes occur simultaneously.

- Assertion** : Mechanical digestion of food involves conversion of complex substances such as carbohydrates into their simple forms such as glucose by enzymatic action.

Reason : Mechanical digestion of food involves breaking down of food by chewing and churning.

- Assertion** : If we talk while eating, food particles may enter into our windpipe and result in choking or coughing.

Reason : Epiglottis which prevents the entry of food into windpipe, may not get closed properly if we talk while eating.

- Assertion** : Large intestine is larger in diameter as well as in length than small intestine.

Reason : Numerous villi are present in the inner wall of small intestine which help it to absorb the maximum amount of nutrients.

- Assertion** : Movements of the wall of food pipe which help to push food down into stomach are referred to as peristalsis.

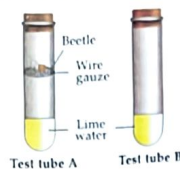
Reason : Peristaltic movements are reversed during vomiting.

► Subjective Questions

Very Short Answer Type

- Name the secretions of inner lining of stomach.
- Why birds do not need teeth?
- What happens when you blow gently into the glass of lime water with the help of straw?

- How does breathing rate of a person change while sleeping and exercising?
- What do you understand by the term mastication?
- In which animals gaseous exchange occurs through lungs?
- Salivary amylase enzyme acts on which food component?
- Write down the percentage of oxygen in inhaled and exhaled air.
- Which is the longest part of alimentary canal?
- What are the parts of human respiratory system?
- Which food components are acted upon by the pancreatic juices?
- What do you understand by the term 'cud'?
- Name the energy giving nutrients.
- What changes take place in the air as it passes through the nasal passage?
- Exchange of gases occurs in which part of the lungs in humans?
- In which form is oxygen carried by blood to the body cells in our body?
- Which organ in human body serves as the common passage for both food and air?
- Name the secretion of liver that helps in digestion of fats.
- Name any three glands of human digestive system.
- Define breathing rate.
- What happens to the volume of chest cavity during exhalation?
- Name different types of teeth found in man.
- Name the two parts of a bird's stomach and explain their functions.
- Write down the role of saliva in digestion process.
- Give one word for the following:
 - The muscular sheet between thoracic cavity and abdominal cavity.
 - Organs of respiration in fish.
 - A gas that turns lime water milky.
 - The respiratory organ in earthworm.
 - Small openings present on sides of bodies of insects for respiration.
- Differentiate between mechanical digestion and chemical digestion.
- How does the food get absorbed?
- What happens to the food in small intestine?
- Study the given experimental set up. In which test tube, will the lime water turn milky after 12 hours and why?



- What are the digestion products of carbohydrates, proteins and fats? In what way are they utilised by human body?
- What happens to the air breathe in, once it reaches the lungs?
- Give one word for the following:
 - The part of stomach that temporarily stores partially chewed food in cows.
 - The semi-digested food in ruminants that is chewed again.
 - The structure that bears taste buds.
 - Cutting and biting teeth.
- How does the nostrils and the nose contribute to the process of breathing?
- Draw a simple diagram showing peristalsis in oesophagus.
- What is rumination?

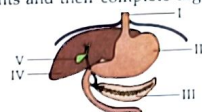
Short Answer Type

- What is the significance of highly coiled nature of small intestine?
 - Absorption of digested food occurs in the stomach. Do you agree?
- How do the gills help in the exchange of gases in fishes?

- Why does air rush into our body when we inhale?
- Give one word for the following:
 - Finger-like projections present in small intestine.
 - Widest organ of alimentary canal.
 - Largest gland of human body.
 - Sac-like structure that stores bile juice.
- Give differences in the composition of air being inhaled and exhaled.
- Mention the different compartments of alimentary canal.
 - Define absorption.
- What is the function of tiny hair and mucus present in nose?
 - State differences between breathing and respiration.
- Draw a neat and labelled diagram of human digestive system.
- Write down the functions of stomach in human beings.

Case Based Questions

Case I: The given diagram shows various parts of human digestive system which contains gastrointestinal tract and the accessory organs that help in the breakdown of food into smaller components and their complete digestion.

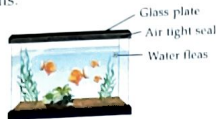


- With the help of simple diagram explain how does digestion of food take place in ruminating mammals?
- Exchange of gases occurs in different ways in different animals. Give examples.
 - How does exchange of gases take place in the lungs in humans?
- With the help of a diagram, explain the structure of human respiratory system.
- How does sneezing or coughing help in the entry of clean air into our body?
 - What is breathing rate (or respiratory rate)? How does it change under different conditions?
- How the food gets digested in different parts of the digestive tract of humans?
- Do all animals possess mouth and anus? Briefly explain with the help of examples.
 - How is food prevented from entering the wind pipe while eating?
- Explain the mechanism of breathing in humans with the help of neat and labelled diagrams.
- Give reasons for the following:
 - Small intestine is 7.5 metres long.
 - Humans cannot digest cellulose whereas ruminants can.
- The labels I, III and IV represent _____, _____ and _____ respectively.
 - duodenum, gall bladder, liver
 - duodenum, caecum, pancreas
 - gall bladder, pancreas, liver
 - oesophagus, pancreas, liver
 - Which among the following enzyme is present in the secretion of II?
 - Trypsin
 - Salivary amylase
 - Pepsin
 - Protease
 - X, Y and Z are found in the part labelled as II. X helps Y to act. Y causes breakdown of proteins. Z protects stomach wall from the action of X. Identify X, Y and Z and select the correct option.
 - X - Pepsin
 - X - Trypsin
 - Z - Mucus
 - Y - HCl
 - Which among the following stores bile?
 - II
 - V
 - IV
 - III
 - The following reaction of digestion of food is given below. Select the option that gives correct location of the given reaction.

A. Carbohydrates → Glucose

- B Fats → Fatty acids
C Proteins → Peptides
(a) A - I (b) B - V
(c) B - III (d) C - II

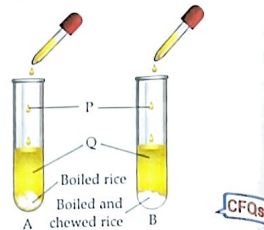
Case II : Different organisms have different ways in which exchange of gases occurs. Different respiratory organs are skin, lungs, gills and some animals even respire through tiny holes. Students were asked to observe the aquarium and identify the method of breathing in aquatic organisms.



- What is required by the organisms to survive in aquarium for longest possible time?
(a) Nitrates (b) Oxygen
(c) Sunlight (d) Carbon dioxide
- How does water enter the body of a fish for respiration?
(a) Through nostrils
(b) Through the mouth
(c) Through the fins
(d) Through the tail
- How water is necessary for survival of organisms in aquarium?
(a) Water provides carbon dioxide for respiration.
(b) Water traps sunlight energy necessary for growth of plants.
(c) Water contains dissolved oxygen necessary for respiration.
(d) Water provides hydrogen gas necessary for respiration.
- Read the given statements and select the correct option.
Statement A : In fishes, gills are devoid of blood supply.
Statement B : Carbon dioxide rich blood is transported to the gills for gaseous exchange.
(a) Both statements A and B are correct.
(b) Both statements A and B are incorrect.

- (c) Statement A is correct but B is incorrect.
(d) Statement A is incorrect but B is correct.
5. Respiratory organ of fish present in aquarium is
(a) spiracle
(b) lungs
(c) buccopharyngeal cavity
(d) gills.

Case III : The given setup shows two test tubes A and B containing boiled rice in A and boiled and chewed rice in B are taken along with Q to check the digestion of complex carbohydrates into simple sugars. To this solution, P is added as shown.



- Along with boiled rice, Q is added in both the test tubes. Identify Q.
(a) Alcohol (b) Acetic acid
(c) Water (d) Acetone
- The given experiment is used to detect the presence of
(a) sucrose (b) cellulose
(c) maltose (d) starch.
- Solution P is used to see the change in colour in the experiment. Identify the solution.
(a) Iodine solution (b) Methylene blue
(c) Fehling's solution (d) Phenolphthalein
- Which enzyme acts on the boiled rice on chewing?
(a) Amylase (b) Protease
(c) Maltase (d) Lipase
- What is the change in colour of solution in test tube B after adding P to it?
(a) Blue-black (b) No change
(c) Light pink (d) Blue-green

Case IV : A group of students are studying the digestive system of birds. Their teacher represented each part of the birds digestive system through a letter.

A: A pouch-like structure where food is stored and softened.

B: The part that secretes digestive juices to begin chemical digestion.

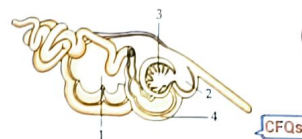
C: A strong, muscular organ that uses grit to grind food.

D: The long, coiled tube that receives food from C.

E: The final part through which undigested food is removed from the body.

- Which of the following is the correct match for part C?
(a) Crop (b) Intestine
(c) Gizzard (d) Cloaca
- What is the function of part D in the bird's digestive system?
(a) Stores food
(b) Grinds food
(c) Absorbs nutrients
(d) Excretes waste
- If part B stops functioning, what will happen?
(a) The bird will not store food.
(b) The bird will not grind food.
(c) The bird will not secrete digestive juices.
(d) None of these.
- What is the correct sequence of the organs through which food passes?
(a) A → C → B → D → E
(b) A → B → C → D → E
(c) B → A → C → D → E
(d) A → B → D → C → E
- Which part (A to E) is most important for a bird's ability to eat quickly and digest later?
(a) A (b) B (c) C (d) D

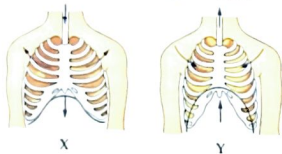
Case V : The given diagram represents alimentary canal of a ruminant which possess a unique digestive system that allows them to better use energy from fibrous plant material than other herbivores. Stomach of these animals consists of four chambers.



- The given alimentary canal is related to
(a) primates (b) elephants
(c) cattle (d) tigers.
- Identify the parts labelled as 1, 2, 3 and 4.
(a) 1 - Rumen, 2 - Omasum, 3 - Reticulum, 4 - Abomasum
(b) 1 - Reticulum, 2 - Abomasum, 3 - Omasum, 4 - Rumen
(c) 1 - Abomasum, 2 - Reticulum, 3 - Omasum, 4 - Rumen
(d) 1 - Rumen, 2 - Reticulum, 3 - Omasum, 4 - Abomasum
- Enzyme produced by the microorganisms harbouring in '1' helps in
(a) digestion of hemicellulose
(b) digestion of pectin
(c) digestion of cellulose
(d) digestion of glycogen.
- Arrange the following steps in the process of digestion of food in ruminants in correct order.
(I) Cud is pushed to '2'.
(II) Food is sent to '3' and '4'.
(III) Partial digestion of food is carried out by anaerobic bacteria that harbour in '1'.
(IV) Cud is brought back into the mouth for chewing.
(V) Food is partially chewed and mixed with saliva.
(VI) Food moves towards the intestine for complete digestion.
(a) III → II → IV → VI → V → I
(b) V → III → I → IV → II → VI
(c) I → IV → II → VI → III → V
(d) III → V → I → IV → VI → II
- Humans cannot digest cellulose like ruminants as
(a) they have longer alimentary canal

- (b) they do not have cellulose digesting bacteria
- (c) they are carnivores
- (d) the enzyme for cellulose digestion is non-functional.

Case VI : During the process of inhalation and exhalation, we observe changes in movements of ribs and diaphragm. The lungs being elastic show a considerable change in size.



- During the process 'X', the diaphragm
 - moves upward
 - moves downward
 - does not move
 - expands sideways.
- Identify the process X and Y.
 - X-Inhalation, Y-Exhalation
 - X-Exhalation, Y-Inhalation
 - X-Expiration, Y-Respiration
 - X-Respiration, Y-Expiration
- Process X and Y involves
 - oxidation of food
 - exchange of gases
 - transport of nutrients
 - release of metabolic wastes.
- What will happen when air is forced out in figure Y?
 - Backward movement of ribs and diaphragm.
 - Diaphragm moves down and ribs move out.
 - No change in ribs but diaphragm moves down.
 - Backward movement of ribs but no change occurs in diaphragm.
- Which of the following does not occur during process 'Y'?
 - Air pressure increases in lungs.

- (b) Diaphragm becomes dome shaped.
- (c) Volume of thoracic cavity increases.
- (d) Ribs move downward and inwards.

Case VII : Rohit took a clean and transparent test tube with narrow opening and poured some freshly prepared lime water in it. He made a hole in the cork and inserted a plastic straw in it. He fixed the cork in the test tube and made it air tight. He then dipped the ends of straw in it air tight. He then dipped the ends of straw in lime water and blew gently through straw into it several times and observed the changes in it.



CFQs

- Lime water turn milky due to the presence of _____ in blown air.
 - oxygen
 - carbon monoxide
 - carbon dioxide
 - nitrogen
- Which organ is majorly involved in this experiment?
 - Brain
 - Lungs
 - Nasal chamber
 - Stomach
- Lime water turns milky due to formation of which of the following products?
 - Calcium carbonate
 - Calcium oxide
 - Calcium sulphate
 - Calcium nitrate
- Which of the following is the chemical formula of lime water?
 - Calcium bicarbonate
 - Calcium oxide
 - Calcium sulphate
 - Calcium hydroxide
- What is the composition of air in the given case?
 - Oxygen (21%)
 - Carbon dioxide (4.4%)
 - Nitrogen (2%)
 - Carbon dioxide (0.03%)



Life Processes in Plants

Learning Outcomes...

How Do Plants Grow?

How Do Plants Get Food for their Growth?

Transport in Plants

Transportation of Water and Minerals

Transportation of Food

Respiration in Plants



Introduction

Green plants are the true "power plants" of our planet. They capture energy from the Sun and use it along with water and carbon dioxide to create organic molecules. This extraordinary ability to convert solar energy into food makes green plants and other autotrophs like algae, phytoplankton and certain bacteria, producers.

Producers or autotrophs are organisms that can make their own food using sunlight, water, carbon dioxide or chemical energy. They perform this through the process known as photosynthesis. All green plants, from tiny mosses to giant fir trees prepare their food through the process of photosynthesis.

How Do Plants Grow?

Observe the different kinds of plants that are present in your locality. They exhibit certain types of changes as they grow over time. As a seed germinates, the shoot emerges from the soil, new leaves are formed, height gradually increases and the stem starts becoming woody. Trees continue to increase their height and girth over a period of time. Plants show various phases of their life during growth, such as vegetative and reproductive phases. There are many life processes that drive plant growth and development, such as photosynthesis, respiration and transpiration, etc.

ACTIVITY CORNER

To study the effect of sunlight and water on plant growth

► **Materials required :** Earthen pots/containers, garden soil, plant saplings, and water.

► **Procedure :** (i) Take three earthen pots/containers filled with garden soil. (ii) Plant saplings of similar sizes of chilli or tomato in each pot.

(iii) Label the three pots as A, B and C.

(iv) Count the number of leaves on each sapling and record your observations.